

MSC ARTIFICIAL INTELLIGENCE  
MASTER THESIS

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# Causal Metrics for Evaluating Deep Learning Climate Emulators

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## Abstract

Traditionally, climate models create physical simulations that are computationally expensive to run. With Deep Learning methods becoming more sophisticated, DL-based “climate emulators” have gained popularity due to their higher accessibility and better computational performance. However, a key problem in evaluating climate emulators remains: identifying whether the emulator has learned a true underlying pattern from the data, or spurious correlations. Currently, emulators are commonly evaluated on their predictive performance on test datasets by calculating the Root Mean Square Error (RMSE) between their prediction and the reference data generated by the traditional climate model being emulated. This is problematic since RMSE is a correspondence metric and does not contain any information about the underlying causal structure learned by the emulator. In this work, we propose to directly compare causal graphs that are inferred from the emulator to the ones underlying reference model. We investigate if different methods of comparing these graphs (causal vs structural metrics) better correlate with model performance. This information is useful to anticipate how well an emulator will perform on out-of-distribution emulation tasks, i.e. if it truly learned dynamics of patterns or if it merely memorized the training data. To achieve this, we use causal representation learning and combined dimensionality reduction and causal discovery methods to infer the underlying latent causal graphs that a climate emulator has learned. We harness the SAVAR stochastic climate model that uses ground truth causal connections between latent modes to generate a synthetic dataset for training, testing, and creating interventions. To learn the graphs, we use the PICABU causal representation learning method, as well as dimensionality reduction in conjunction with the Multidata-PCMCI+ causal discovery method. We contrast performance of both methods by assessing how well they recover the ground-truth latent causal graph from the data generated by SAVAR. We investigate the relationship between the structural graph comparison metrics F1 score and Structural Hamming Distance (SHD), the causal graph comparison metric Parent-AID, and the model performance comparison metrics RMSE, RMSE under intervention, and Least Squares Distance (LSD) between Fourier transform coefficients. We find that structural graph comparison methods are not predictive of model performance while causal comparison methods may be. Specifically, we find that Parent-AID has a positive correlation with model RMSE and LSD across various emulators (MLP, CNN, LSTM, VAE) we trained. Parent-AID has a moderate-to-strong negative correlation with RMSE under intervention. These findings present early evidence that causal graph metrics are a better way to assess a climate emulator’s “understanding” of climate patterns in the data.

Code is available at the following GitHub repository <https://github.com/RolnickLab/causal-graph-comparison>.

# Chapter 1

## Introduction

Deep learning-based climate emulators (hereafter ‘DL emulators’) have shown that they can significantly accelerate climate projections from Earth System Models (ESMs) [13, 43, 75, 78]. They do so by imitating the behaviour of ESMs without having to solve the same systems of computationally prohibitive differential equations [1, 17]. The efficiency gained can enable tasks that would otherwise be inaccessible at scale, such as counterfactual experiments and large ensemble projections. However, a key challenge for climate emulators is to create stable realizations of climate dynamics that are consistent with the dynamics of the ESM on which the emulator was trained [3][18]. The adoption of emulators by climate scientists and policymakers is thus contingent on their credibility, robustness, and trustworthiness, which can be ameliorated through robust evaluation [75].

Emulators are commonly evaluated using variations of Root Mean Square Error (RMSE) or Mean Absolute Error (MAE), which quantify the difference between projections from an emulator and its reference ESM [9, 13, 78, 80]. However, such regression-based metrics are insufficient for guaranteeing that the emulator has not simply learned spurious correlations, and do not address stability and consistency issues. Instead, metrics are needed to evaluate whether emulators capture the underlying climate dynamics of the emulated ESM, not just whether targets and outputs align [75].

One way to compare the dynamics of two systems is to directly compare their differential equations. In practice, however, we cannot recover differential equations from the outputs of a DL-based climate emulator. We thus propose to compare models based on the causal relations that we can learn on their output data (projections). These cause-and-effect relationships between high-level variables in model projections can be identified using dimensionality reduction combined with causal discovery, or by using causal representation learning algorithms. Their graph representations can then be compared between models using graph comparison metrics [63]. The underlying assumption here is that the more similar the causal graphs are between the emulator and the ESM data, the “better” the emulator has captured the dynamics of the reference ESM (the “target”).

We train a set of autoregressive emulators that each capture different aspects of the data, including non-linearities, time connections, spatial information, and latent representations. These models are trained on SAVAR, a simplified climate model that generates spatial time-series data with known causal relationships [74]. We compare two causal inference methods: (1) the first combines dimensionality reduction via weighted averaging with causal discovery using PCMCI+ [64] and (2) the second method employs causal representation learning through PICABU [25]. These methods are used to learn time-lagged latent causal graphs on both the emulator outputs, and SAVAR data. The resulting graphs are then compared.

Existing literature that compares causal graphs in climate models primarily measures similarity through structural differences, using metrics such as F1 score applied to edge identification, and Structural Hamming Distance (SHD) [10, 14, 30, 47, 74, 51]. Structurally similar graphs, however, can have very different causal implications [51], which is why metrics considering causal effects between variables might be better suited for this task. This includes (1) Adjustment Intervention Distance (AID) [24], which compares causal effects across interventions; and (2) distance average causal effect [58], which considers link weights to compare the causal importance of each node between two graphs. A main contribution of this work

is to understand whether the causal effect-based metrics discussed above are better suited for evaluating dynamics learned by DL-based climate emulators than structure-based metrics.

To validate whether our proposed metrics are indeed useful for measuring how well causal dynamics have been captured, we assess whether they are predictive of “good” model performance, benchmarked by RMSE and analysis of power spectral density [56]. We then evaluate the performance of these metrics under different known intervention scenarios. Intervening on the SAVAR graphs [15] provides us with a tool to validate whether an emulator has learned robust causal dynamics. We hypothesize that the causal graph of an emulator that has learned the right dynamics will have better projection performance on intervention data, measured by RMSE. The question we seek to address is whether evaluation using causal metrics pre-intervention can be more predictive of this robustness as opposed to structural-based metrics. This study applies a causally-informed evaluation framework for assessing fidelity of internal causal dynamics learned by ML emulators. The method was introduced by Nowack et al. [47] for evaluating numerical climate models. Consequently, this work applies a causal metric for comparing causal graphs, and compares its performance to other commonly used structural metrics. We find strong correlation between the causal metrics (Parent-AID/OSet-AID) and model performance (RMSE/RMSE under intervention) and very weak correlation between structural metrics and model performance. This indicates that using a causal representation learning framework in combination with a causal metric provides more robust insights into whether a climate emulator has accurately captured climate dynamics.

In chapter 2, we review the background on climate science and causality, present a user study that helped guide our research questions, and discuss related works. In chapter 3, we present the methodology and technical implementation details of the experiments. In chapter 4, we present the results and figures. In chapter 5, we summarize our contributions and discuss opportunities for future work.

# Chapter 2

## Background

This background chapter introduces key concepts in climate science relevant to numerical physics-based climate models and Deep-Learning based climate emulators. It then presents insights from a user study that was essential in refining the research focus, followed by causality fundamentals and methods central to this thesis, including as causal discovery, representation learning, and graph comparison techniques. This chapter concludes with a literature review of related works directly relevant to the methods used in this thesis.

### 2.1 Climate Science

This section introduces the climate science background that forms the context for this work. We begin by explaining what numerical climate models are, how they work, and why DL emulators have emerged as an important complementary approach. Understanding current evaluation practices and their limitations is crucial for developing more robust methods that can reliably assess DL emulators.

#### 2.1.1 Earth System Models

Numerical Earth System Models (ESMs) simulate the complex interactions within and between the atmosphere, ocean, land, and sea ice, by representing physical, chemical, and biological processes as mathematical equations [19, 79]. These equations describe how climate variables, such as pressure, temperature, and precipitation, evolve and interact over time, enabling projections about potential futures of the Earth system [40]. These systems of equations respond to “forcings”, which are factors that can change the balance of energy in the Earth system and influence the climate. Forcings can be anthropogenic, in the case of greenhouse gas (GHG) emissions, or natural, in the case of wildfires [69]. There are many plausible pathways that describe how forcing concentrations will change over time, influenced by drivers such as human lifestyle, technology use, and policies. The Shared Socioeconomic Pathways (SSPs) framework is a widely recognized set of scenarios that represent distinct narratives of future socioeconomic developments and associated forcing concentrations [57]. By running climate models on SSP scenarios, researchers investigate how different future trajectories of forcings such as carbon dioxide (CO<sub>2</sub>) could impact the climate, providing a scientific basis for informing policymakers, who rely on these scientific reports for preparation, adaptation, and mitigation of climate change. One example is the “Coupled Model Intercomparison Projects” (CMIP) [1], containing data obtained from various ESMs, with comparisons and evaluations used in the reports by the Intergovernmental Panel on Climate Change (IPCC) [19].

By explicitly representing fundamental laws of physics in their model of the world, numerical ESMs have a high degree of fidelity and robustness in simulating the Earth’s climate [13, 75]. Although these systems of partial differential equations (PDEs) allow scientists to reason over model behaviour, they are often prohibitively complex to solve directly [20]. Numerical ESMs thus approximate solutions by discretizing the globe into grid cells, and representing smaller “subgrid” processes that cannot be captured at the chosen resolution, such as cloud formation, using simplified parameterizations. Even with these approximations, however, a single simulation can require months to compute [40][1][20]. The computational cost of numerical ESMs limits both the number and resolutions of scenarios that scientists can explore [78]. Although coarser spatial and temporal resolution can allow for more efficient simulations, the trade-off

in these design choices limits the model’s ability to represent detail at a level that could be impactful for local communities facing climate change impacts. Furthermore, limited available runs restrict our ability to, among others, (a) distinguish between changes driven by external forcings and those arising from natural variability, (b) benefit from large ensembles that can enable tasks such as mitigating individual model biases, improving uncertainty quantification, (c) and capture rare events, (d) conduct probabilistic and counterfactual analyses, and (e) equip policy makers with a diverse range of projections that explore different climate change mitigation pathways. [17, 75]

### 2.1.2 Climate variability

Climate variability arises from internal “noise” within the Earth system, which produces fluctuations that occur independently of external forcings, and influences weather and climate globally over time scales that can range from days to decades [47, 69]. One prominent mode or driver of variability, for example, is the El Niño-Southern Oscillation (ENSO), a large-scale climate pattern that emerges from complex interactions between the ocean and the atmosphere in the tropical Pacific [13]. Through [teleconnections](#), which are atmospheric or oceanic pathways that link changes in climate variables across distant regions, ENSO affects temperature and precipitation across the globe in interannual cycles [72]. The term “climate indices” refers to well-documented modes of variability, such as ENSO, that represent a subset of the full spectrum of climate dynamics. Beyond affecting regional climate variables, these modes can in turn interact with one another to amplify or dampen global or regional effects. Building models that faithfully reproduce the climate system’s interconnected dynamics would thus benefit from understanding the causal relationships between modes of variability. However, discovering these causal structures is challenging as we can neither perform controlled experiments on the climate system nor observe multiple realizations of Earth’s climate to disentangle cause from correlation. See Section 2.3 for a detailed discussion of causality.

### 2.1.3 SAVAR climate model

The Spatially Aggregated Vector-AutoRegressive (SAVAR) model is a simplified climate model that emulates the interactions of variables on a spatio-temporal grid [74]. This section briefly introduces SAVAR, which will be used to generate simulated data for this study. As discussed in Section 2.1.2, real-world ground truth (GT) causal relationships between climate phenomena are unknown and are difficult to discover and validate. SAVAR provides a “sandbox” environment where the true causal graph dictating dynamics is known, enabling benchmarking of causal discovery methods for climate science. Although GT causal relationships could in theory be derived from the PDEs in more complex numerical ESMs, the high level of complexity renders the task intractable. Further discussion on causal discovery can be found in Section 2.3.2.

In SAVAR, the value of each grid-level variable is linearly dependent on the value it and other variables had at previous discrete time steps. The model does not include external forcings, meaning that all dynamics emerge from interactions within the system. The temporal evolution of the variables are generated through a vector autoregressive (VAR) process. The climate variables are spatially aggregated into non-overlapping modes. Variables are governed by “modes” which act as [latent variables](#) that drive grid-level behaviour. For further discussion on latent variables, see Section 2.3.3. The grid-level variables are aggregated into non-overlapping 10×10 Gaussian clusters (“modes”). For example, Fig. 3.3 shows data generated by four modes, visible as Gaussian patches of different hues in the four quadrants of the grid. Each mode has “time-lagged causal links”, meaning that it may be influenced by itself or other modes from the past. Since SAVAR is defined by a known structural causal model, we can directly access the ground truth causal graph that represents the relationships between modes (see Section 2.3.1). We also have access to the mapping from the modes to the grid-level variables. **SAVAR thus generates spatial time series using a known causal graph over the latent modes.** For

simplicity, SAVAR assumes a flat, regularly spaced grid and models a single unit-less variable. In contrast, a more comprehensive climate model would account for land–sea delineations, curvature of the Earth, multiple interdependent variables, overlapping modes of variability, and three-dimensional spatial structure that includes elevation. Despite these simplifications, SAVAR offers a useful framework for applying and evaluating causal inference techniques in climate science.

#### 2.1.4 Deep Learning-Based Climate Emulators

The previous section 2.1.1 highlighted the computational limitations of numerical ESMs. Emulators have emerged as a way to address these scaling challenges by approximating the behaviour of numerical ESMs without explicitly solving the complex systems of physical equations underlying them [13, 43, 75, 78]. Broadly speaking, the emulator is trained to minimize the discrepancy between its own predictions and the predictions of the ESM, given the same input, in order to emulate the behaviour of the ESM for various projection tasks at inference [9][80].

Emulators can either: (1) be simplified versions of their numerical counterparts (“physics-based”), (2) approach the task as an *inverse* problem, using statistical methods to infer the model from data (“data-driven”) [73], or (3) combine physical knowledge with data (“hybrid”). DL emulators are a further sub-category of statistical methods for emulation that use neural networks to learn mappings between inputs and outputs of ESMs [78]. Neural network architectures for DL emulators include spherical Fourier neural operators (SFNOs) [5, 9, 80] and U-Nets [29]. This work focuses on DL emulators that explicitly aim to replicate the full behaviour of numerical ESMs. DL-based “hybrid” models do not fully fit our working definition of a full DL-based climate emulator for this work [46, 78]. Other non-DL data-driven methods such as linear pattern scaling [39] are also outside of the scope of this work.

By speeding up climate projections, emulators enable the exploration of a wide range of future scenarios computationally inaccessible with numerical ESMs [78]. The benefits include building large ensembles of projections to characterize uncertainty, and allowing climate scientists as well as policymakers to quickly investigate the impact of policies or better understand the drivers of observed climate events. However, since any underlying physical dynamics are implicitly learned [11], a key challenge for climate emulators is to create stable realizations of climate dynamics that are both internally consistent, and consistent with the dynamics of the ESM on which the emulator was trained [3][18]. The adoption of emulators by climate scientists and policymakers is thus contingent on their credibility, robustness, and trustworthiness, which can be ameliorated through robust evaluation [75].

#### 2.1.5 Evaluating Deep Learning-Based Climate Emulators

DL emulators often function like “black boxes”, unlike ESMs which allow for component-level testing and validation [75]. DL emulators learn statistical correlations without explicitly representing different physical processes as distinct, so it is difficult to verify whether they correctly learn the dynamics of the ESM being emulated [75]. This makes it difficult to trust that DL emulators will generalize to out-of-distribution climate change scenarios, thus undermining their utility for future planning. As Ullrich et al. [75] emphasize, “[...] they need to get the right answer for the right reasons for understanding and simulating future or past climates.” Building confidence in emulators thus requires robust evaluation.

Numerical ESMs are assessed by comparing to historical observations and trends (either measured or simulated through reanalysis data), monthly averages, documented modes of variability (such as ENSO), emergent relationships, and theoretical constraints [75]. It is important to emphasize that the goal of evaluating full DL emulators in the context of this work is not to surpass performance of the ESM being emulated - emulators are instead evaluated by how closely they reproduce ESM projections. Evaluating DL models using observed and reanalysis data is outside of the scope of this work.

Emulators are commonly evaluated using summary metrics such as Root Mean Square Error (RMSE) or Mean Absolute Error (MAE) to quantify discrepancies between emulator predictions and ESM projections [9, 13, 78, 80]. Regression-based metrics, however, are insufficient for guaranteeing that the emulator has learned correct dynamics rather than spurious correlations [63]. Moreover, due to the climate’s internal variability (see Section 2.1.2) a degree of uncertainty is always present in climate simulations, meaning perfectly accurate deterministic predictions are unattainable [69][19]. Even if they were, perfect alignment is not desirable as it may indicate overfitting, reducing the emulator’s ability to generalize. There are probabilistic DL emulators that can quantify uncertainty, but even these rely on mean statistics that encourage “smooth” predictions, missing high frequency variability, rare events, and localized extremes. On the other hand, “noisy” models with higher variability also face their own challenges, risking amplifying noise in long autoregressive rollouts. Instead, metrics are needed to evaluate whether emulators capture the underlying climate dynamics of the emulated ESM, not just whether targets and outputs align on average [75].

The following provides a description of prevalent evaluation metrics for DL emulators. Basic methods such as bias, spread, and correlation are also common but not described below. Many of the below described metrics are available in packages such as the PCMDI Metrics Package (PMP) [37] or ESMValTool [16].

### Deterministic metrics

**Root Mean Square Error (RMSE, [45, 75]):** Represents the standard deviation of the difference between the model prediction and the target values for deterministic forecasts. A value of 0 indicates perfect alignment. RMSE suffers from double penalty, which means if a particular value is predicted at the wrong spatial or temporal displacement, a penalty is counted once for failing to appear where expected, and once for appearing in the wrong “location”. Additionally, RMSE encourages a “smoothing effect” by penalizing outliers, thus estimates are less able to reproduce variability, remaining closer to the mean. For DL emulators, variations of RMSE are often used. For example, the DL emulator ACE2 [80] adjusts for latitude differences to ensure proportional contribution of each grid cell:

$$\text{RMSE} = \sqrt{\sum_{\phi, \lambda} w_{\phi, \lambda} (\overline{y(\phi, \lambda)} - \hat{y}(\phi, \lambda))^2} \quad (2.1)$$

Where:

$\phi, \lambda$  Represent latitude and longitude, respectively.

$w_{\phi, \lambda}$  Area weight for a particular grid cell  $(\phi, \lambda)$ , with  $\sum_{\phi, \lambda} w_{\phi, \lambda} = 1$ .

$\hat{y}(\phi, \lambda)$  The predicted variable at  $(\phi, \lambda)$ .

$\overline{y}$  the target quantity, with the overline representing the average over a period of time.

**Mean Absolute Error (MAE, [18]):** MAE is similar to RMSE in that it measures average prediction error, but is less sensitive to outliers. This can make it a better choice for training objectives if preserving variability is important.

**Anomaly Correlation Coefficient (ACC, [45]):** When an emulator makes a prediction that deviates from the mean, ACC is a measure of whether this anomaly is correlated with spatially and temporally with anomalies produced in the reference ESM. Here, “mean” refers to [climatology](#), the long-term average of a climate variable at a particular time of year.

**R-squared, ( $R^2$ , [45]):** Addresses the limitations of RMSE by providing a measure for how well a model reproduces both long-term average behaviour and short-term variability [80].

## Performance metrics

**Power Spectra** [45, 56]: Power Spectral Density (PSD) represents a time series by its frequency components. For climate data, low frequencies reveal long-term climate trends and cycles, and high frequencies reflect short-term weather variability. The Spectral Residual metric is a “physics-based” metric that measures the RMSE between all frequencies of the spectra for the prediction and target forecasts. Unlike simpler error metrics such as RMSE which only measure point-wise accuracy, spectral metrics evaluate whether forecasts preserve natural climate oscillations (like El Niño cycles) and maintain realistic variability across time scales.

**Emergent constraints and sanity checks** [75]: Assess whether model behaviour is consistent with physical principles, such as conservation of mass, or how much humidity air can hold before it becomes precipitation.

**Covariance** [75]: Tests model consistency by measuring whether variables vary together in expected ways. Although valuable, this is very difficult to implement across a broad range of variables.

## Probabilistic metrics

**Continued Ranked Probability Score (CRPS, [45])**: Evaluates how closely the distribution of a probabilistic forecast matches that of the reference target. CRPS is similar to MAE in the deterministic setting. A value of 0 indicates perfect alignment.

## Distance-based metrics

**Weighted Wasserstein distance (WWD, [58])**: Measures the “work” needed to transform the emulator’s probability distribution of sea surface temperatures (SST) into the target distribution. Each region is assigned a weight that reflects its connectivity strength, which is the sum of the covariances it shares with all the other regions of the network.

## Comparing Dynamical Systems

While dynamical system-based metrics lack interpretability and are remain an active area of research, it is valuable to consider evaluation methods that extend beyond common summary metrics discussed above.

**Instantaneous Dimension (ID [18])**: Quantifies system complexity at any given moment, measuring the degrees of freedom available to the system’s trajectory. Predictable systems exhibit low instantaneous dimension, while chaotic systems show high values.

**Inverse Persistence (IP[18])**: Characterizes how long a system remains in particular state-space regions before transitioning. High persistence indicates “sticky” states, while low persistence suggests rapid state transitions. These indices can reveal whether machine learning models capture appropriate dynamical complexity and state transition behaviours. Though these dynamical indices (IP and ID) offer valuable quantification of model behaviour, they do not reveal the internal representation of physical climate dynamics that a model has learned. Furthermore, these indices require long forecasts to work properly, and they are not sufficient as diagnostic tools to identify why errors arise.

**Koopman Operators**: Provide a framework for analyzing nonlinear dynamical systems through linear function mappings. However, interpreting Koopman operator properties in terms of real-world phenomena remains challenging, and direct comparison of different systems is an ongoing research problem [34, 4].

Table 2.1: User Study Participants

| Name                         | Role               | Affiliation                        |
|------------------------------|--------------------|------------------------------------|
| Sebastian Hickman            | PhD Student        | University of Cambridge            |
| Prof. Duncan Watson-Parris   | Assistant Prof.    | University of California San Diego |
| Prof. Christian Lessig       | Professor          | OvG-Universität Magdeburg          |
| Prof. Peer Nowack            | Professor          | Karlsruhe Institute of Technology  |
| Prof. Pierre Gentine         | Professor          | Columbia University                |
| Dr. Oliver Watt-Meyer        | Lead Research Sci. | Allen Institute for AI             |
| Dr. Marie McGraw             | Research Sci.      | Colorado State University          |
| Dr. Fernando Iglesias Suarez | Research Sci.      | Institute for Atmospheric Physics  |
| Prof. Tom Beucler            | Assistant Prof.    | University of Lausanne             |
| Dr. Fangfei Lan              | Postdoctoral Res.  | University of Lausanne             |
| Prof. Gianmarco Mengaldo     | Assistant Prof.    | National University of Singapore   |

## 2.2 User Study: Guiding our research focus

As discussed in the previous section, the research landscape for evaluating DL emulators is broad. In order to gain a better understanding of the current state, challenges, and actual needs in this research domain, we conducted a user study featuring ten forty-five minute interviews with domain experts from March to May 2025. Participants were selected based on their expertise in climate science, machine learning or causality. This study does not feature a full qualitative analysis of the interviews, instead we highlight key overall takeaways that help inform our research questions and methodology for this work. This summary is synthesized from recordings and notes taken during the interviews. Interviews were conducted by Christina Isaicu (University of Amsterdam, Mila - Quebec AI Institute) and Julia Kaltenborn (PhD student, McGill University, Mila - Quebec AI Institute). Table 2.1 lists the eleven interviewees who participated in this study. The interviewees were asked the following questions:

1. What are the metrics you use to evaluate the models you have been working with recently?
2. Why are these specific metrics or methods used?
3. Are there any limitations to these approaches, and have you run into any issues?
4. Is there any “magical metric” you wish existed, and if so, why does it not already exist?
5. What are the challenges, if any, in collaborating between climate scientists and machine learning researchers?

These questions were designed to identify gaps in current DL emulator evaluation practices and understand the kinds of evaluation tools that researchers would most need and value. We approach these interviews with the broad question, “How can we better evaluate DL emulators?”, and allow participant feedback to shape our research focus to ensure that our efforts align with real needs in the field. We identify three broad categories of topics pertaining to our investigation: (1) the meta goals that metrics should address, (2) priorities for evaluating DL emulators and, (3) challenges in causal evaluation for DL emulators.

### Meta goals

Interviews emphasized that trust in DL emulators is currently lacking but essential for real-world applicability in areas such as climate policy decisions. Ideally, evaluation metrics should help build this trust by providing interpretable, reliable assessments of emulator performance. Every participant stresses that no “gold standard” metric exists. Each metric reduces the dimensionality of model outputs to a single number, so a single metric is limited in what it

reveals in isolation. It became clear that rather than seeking a novel general-purpose metric, a more valuable contribution of this work would involve developing metrics tailored to specific phenomena or improving underdeveloped evaluation areas. However, participants highlighted a tension between comprehensiveness and usability. One noted that large scorecards make it difficult to compare models, however another participant observed that reaching consensus among climate modelers on any single skill metric would be challenging. Regardless, simplicity, ease of use, and interpretability emerges as essential qualities for metrics that would ensure adoption. Several participants (4/10) also expressed interest in diagnostic capabilities—metrics that identify not just errors, but why they occurred, but they also acknowledged this as a non-trivial challenge. Ultimately to increase our trust in emulator performance we need metrics that assess multiple dimensions and are easy to use and understand.

## Evaluation priorities

All ten interviews emphasized the importance of ensuring adherence to physical laws and constraints. Other priorities include: assessing the fidelity of climate dynamics (consistency in spatial and temporal patterns, capturing realistic trends and internal variability, etc.), evaluating co-variance of variables, assessing whether the emulator outputs the same marginal and joint distribution of variables as the reference model, and measuring “distance” from the reference model. Participants also highlighted the importance of understanding which sources contribute to uncertainty in emulator estimates, whether they arise from internal variability, model structure, parameters, or scenarios (“uncertainty estimation”). One participant uniquely emphasized the importance of data management, noting that preprocessing and augmentation choices can have significant impacts on results. Finally, a key requirement is ensuring that emulators exhibit appropriate causal responses to interventions, meaning that changing the value of a driving variable should produce expected downstream effects.

## Causal evaluation

Participants identified several specific causal evaluation goals: handling non-stationary, multi-scale, and non-linear causal relationships; addressing cyclicity and interdependence in climate variables; assessing interventions; and comparing causal graphs (including temporal lags) and causal functions (strength of the causal connections). Three interviews highlighted an important limitation in current causal methods applied to climate models: To render causal discovery methods tractable, practitioners typically perform dimensionality reduction to reduce the number of variables. However, the choice and intensity of the reduction needs to be carefully selected to strike a balance. Too little reduction leads to intractability, while too much can produce oversimplified causal networks that lack utility.

It became apparent that instead of developing a universal metric, our contributions would be more impactful by addressing evaluation challenges from a focused perspective. A recurring theme we identified involves determining whether models correctly capture dynamics between variables. The application of causality, which examines cause-and-effect relationships, is growing in climate science as a means to address these questions. We choose to focus on causal model evaluation for three reasons: (1) available expertise within our research group, (2) the ongoing development of causal evaluation methods for DL emulators in which improvements are still called for, and (3) the potential in using causality for process-oriented model evaluation that goes beyond standard regression methods. Despite methodological challenges, we identified that comparing DL emulators to their reference model using causal graphs and causal functions offers a promising approach with well-defined opportunities for advancement. Before describing the methods employed in this work, it is essential to first provide a brief overview of causality.

## 2.3 Causality

Causality is broadly concerned with how events, or variables, influence one another. In the context of climate science one could ask, for instance, if the temperature in region A increases, will it affect the temperature in region B? More importantly, how do we know that a temperature change in region B was caused by changes in region A, rather than temperature changes in both regions being correlations in the observed data, driven by a common cause or confounder influencing both regions at the same time?

Causal inference provides the tools for determining causal effects from observed data, moving beyond statistical correlations. With the recent developments in ML models, and the growing availability of observations, data-driven climate models are becoming key for fast exploration of the climate system under unseen emission scenarios. However, to ensure that we can trust these models for out-of-distribution predictions, it is crucial to ensure that they actually learn the causal connections between the variables of the system, rather than relying on correlations in the observed data.

### 2.3.1 Preliminaries

The following section provides a brief overview of important concepts in causality that are relevant for understanding the methods used in this work. Unless stated otherwise, the concepts described in this section are not formal definitions. These preliminaries draw heavily from the following sources: Jewell et al. [27], Spirtes et al. [71], Pearl [48].

#### Probabilistic Dependencies

The probability of a random variable  $X$  describes the likelihood of that variable taking on a particular value, e.g.,  $P(X = x)$ . A central notion in causality is dependence between variables. Two variables are independent if knowledge about one provides no additional information about the other, i.e.,  $P(A | B) = P(A)$ . Otherwise, they are dependent. Two dependent variables may become independent when conditioning on a third variable (or set of variables), referred to as conditional independence, i.e.,  $P(A | B, C) = P(A | C)$ . If the conditioning set that renders  $A$  and  $B$  conditionally independent is empty, this is referred to as marginal independence where  $P(A, B) = P(A)P(B)$ .

#### Causal Graphs

We can visualize the causal relationships between a set of variables as a directed acyclic graph (DAG). A DAG  $G = (V, E)$  has vertices  $V$  that represent the variables, and directed edges  $E$  that encode causal influence. The direction of an arrow  $X \rightarrow Y$  implies causal influence, i.e., that a change in  $X$  can cause a change in  $Y$ . The DAG is acyclic, meaning that there are no directed paths from a variable  $X$  that loop back to itself.

#### Properties and assumptions of DAGs

DAGs follow a few properties and assumptions. The Causal Markov Property of DAGs states that each variable is independent of its non-descendants (non-downstream variables) when conditioned on its parents (direct causes), which allows the joint probability distribution over all variables to be factorized in a way that reflects the graph structure, representing the probability distribution of all variables.

$$P(\mathbf{x}) = P(x_1, \dots, x_d) = \prod_{i=1}^d P(x_i | \text{pa}(x_i))$$

Where  $P(\mathbf{x})$  is the joint distribution of  $d$  random variables, and  $\text{pa}(x_i)$  denotes the set of parents for variable  $x_i$  in the causal graph.

$$X \longrightarrow Z \longrightarrow Y$$

Figure 2.1: A simple directed acyclic graph (DAG) where  $X$  is a parent of  $Z$ , and  $Z$  is a parent of  $Y$ . This is the graphical representation of the following probability distribution  $P(X, Y, Z) = P(X) P(Z | X) P(Y | Z)$

Two DAGs are Markov equivalent if they both have the same set of conditional independencies, or d-separations (the “d” stands for directional). Graphs in this Markov equivalence class (MEC) may have different link orientations, which cannot be differentiated from observational data alone. Graphically, this means that they share the same “skeleton” graph with undirected edges, and they have the same **v-structure**. The directed edges are must follow Meek rules for orientation [41]. Complete Partial Directed Acyclic Graphs (CPDAGs) are graphs with directed and undirected edges that represent DAGs in an MEC. Undirected edges in a CPDAG represent where the DAGs differ. Conditional independencies among the variables can also be inferred directly from the graph, by using the graphical criterion d-separation.

Before describing d-separation, we must first understand two categories of nodes: colliders and non-colliders. A collider on a path in a DAG is a node where both edges point towards it. For example,  $Z$  is a collider on the path  $X \rightarrow Z \leftarrow Y$ . This is also called a **v-structure** if the collider’s parents are not adjacent. A non-collider may have both edges pointing outwards  $X \leftarrow Z \rightarrow Y$  (referred to as a fork), or one incoming and one outgoing edge  $X \rightarrow Z \rightarrow Y$  (referred to as a link).

For two variables  $X$  and  $Y$  to be d-separated, or conditionally independent, every path between  $X$  and  $Y$  must be “blocked” by a given set  $Z$ . If  $Z$  does not block every path, the two variables are conditionally dependent given  $Z$ . A path can be blocked by the following types of nodes: (1) A collider or its descendants that is not in the conditioning set  $Z$ ; (2) A non-collider that is present in the conditioning set  $Z$ . The relevance is that we can translate back and forth between probability distributions and DAGs.

The faithfulness assumption ensures that a causal graph reflects all independencies in the data, and vice versa. This assumption implies that there are no coincidental independencies in the data. Causal sufficiency assumes that all confounders of the relevant variables have been measured (observed) in the data. This way we can trust that if there is a link between  $X$  and  $Y$ , it is a “true” link and not one that is confounded by an unobserved variable. In practice, this assumption is often not realistic to make, so it is important to be cognizant and transparent about the limitations and assumptions of any given dataset.

**CPDAGs** are graphs that represent a Markov Equivalence class of DAGs, containing both directed and undirected edges. An undirected edge represents that the edge could point in either direction, and both resulting graphs would still have the same conditional independencies.

## Adjacency matrices

An adjacency matrix  $A_{n \times n}$  can represent a DAG  $G$  in matrix form, where  $n$  is the number of variables in  $G$ . A non-zero value in element  $A_{ij}$  represents a directed edge from variable  $i$  to variable  $j$ . The matrix can be binary (reflecting only whether there is a link) or weighted (reflecting the value or strength of the link, as well as its presence). A temporal adjacency matrix has an additional dimension which represents temporal lag, e.g., (parent  $\times$  child  $\times$  time), (time  $\times$  parent  $\times$  child), etc., where time is ordered sequentially from  $t - \tau$  ( $\tau$  being the maximum time lag) through  $t - 1$  to the current time step  $t_0$  (from past to present). A flattened adjacency matrix is of size  $n(\tau + 1)$ . Each variable at a particular time step is treated as unique in this graph. The flattened matrix has a crucial temporal causal constraint: links are only allowed forward in time between variables, so a variable at  $t$  cannot affect any of the preceding time steps. For the purpose of this work, we assume there are also no instantaneous links, meaning a variable cannot affect itself nor any other at the same time step.

To illustrate valid links, the matrix below shows the flattened adjacency matrix for a fully connected graph for one time step and three variables (i.e., the temporal adjacency matrix is all ones). The matrix exhibits a block lower-triangular pattern. Furthermore, if a link is present from  $t - 1$  to  $t_0$ , but  $\tau_{\max} = 5$ , then in the flattened adjacency matrix the link is also present from  $t - 2$  to  $t - 1$ , all the way to  $t - 5$  to  $t - 4$ . Since the dimensionality of this graph is quite large, it would otherwise be incredibly sparse. This will help with calculations down the line.

A flattened adjacency matrix  $A \in \{0, 1\}^{n(\tau+1) \times n(\tau+1)}$  is defined by

$$A_{(i,t),(j,t')} = \begin{cases} 1, & \text{if there is a directed edge } X_i^{(t)} \rightarrow X_j^{(t')} \text{ and } t < t', \\ 0, & \text{otherwise.} \end{cases}$$

For example, for  $n = 3$  variables and  $\tau = 1$  (two time steps,  $-1$  and  $0$ ), a fully connected flattened temporal graph has the following structure. The edge direction is from row  $\rightarrow$  column, from parent to child, respectively.

$$A = \begin{matrix} & X_{1,t-1} & X_{2,t-1} & X_{3,t-1} & X_{1,t} & X_{2,t} & X_{3,t} \\ \begin{matrix} X_{1,t-1} \\ X_{2,t-1} \\ X_{3,t-1} \\ X_{1,t} \\ X_{2,t} \\ X_{3,t} \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 0 \end{bmatrix} \end{matrix}$$

## Structural causal models

Structural causal models (SCMs) are another way of representing causal relationships within a system of variables. There are two types of variables in an SCM: exogenous variables that are external to the model, where we do not need to know how they came about; and endogenous variables, which are internal to the system and are influenced by at least one exogenous variable. The relationship between these variables is reflected by a set of equations modelled by functions of direct causes, with added error drawn from a particular distribution to account for stochasticity. Each equation represents a distinct causal relationship between variables. This section draws from Bongers et al. [6].

A variable  $X$  is a direct cause of a variable  $Y$  if  $X$  appears in the function that assigns  $Y$ 's value.  $X$  is a cause of  $Y$  if it is a direct cause of  $Y$ , or of any cause of  $Y$ . An SCM is formally defined as  $\mathcal{S} = (U, V, F)$  where  $U$  denotes the set of exogenous variables,  $V$  the endogenous variables, and  $F$  represents the set of functions.

For example, let  $\mathcal{S} = (U, V, F)$  be an SCM defined by

$$U = \{\epsilon_X, \epsilon_Y, \epsilon_Z\}, \quad V = \{X, Y, Z\}, \quad F = \{f_Z\},$$

with

$$\begin{cases} X \leftarrow \epsilon_X \\ Y \leftarrow \epsilon_Y \\ Z \leftarrow 2X - Y + \epsilon_Z \end{cases} \quad \epsilon_X, \epsilon_Y, \epsilon_Z \sim \mathcal{N}(0, 1) \quad (2.2)$$

In this model, the exogenous variables in  $U$  are each drawn from a normal distribution and are independent of one another.  $X$  and  $Y$  are independent root causes, and their values are set to their respective noise terms  $\epsilon_X$  and  $\epsilon_Y$ .  $Z$  is a linear combination of  $X$  and  $Y$  through the function  $f_Z(X, Y) = 2X - Y$  plus its own noise term  $\epsilon_Z$ . More generally, each endogenous variable  $X_i$  for  $i \in V$  can be expressed as a function of its parents in  $G$  and a noise term  $\epsilon_i$ :

$$X_i \leftarrow f_i(X_{\text{Pa}(i)}, \epsilon_i).$$

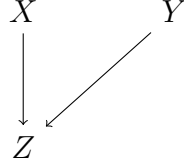


Figure 2.2: The above SCM described in Eq. 2.2, where  $X$  and  $Y$  are direct causes of  $Z$ , can be represented graphically as a DAG.

## Interventions

The following two sections (Interventions and Adjustment Sets) are directly relevant to understanding the causal metrics (Adjustment Identification Distance [24]) used in this work (see Sec.2.4). The purpose of interventions is to observe how other variables are affected when the value or function of a particular variable in a system is changed. SCM notation provides a way to represent interventions as the modification of relevant structural equations. An intervention can be “hard”, where the value is fixed to a constant, or “soft”, where the function – that defines how a variable depends on its parents – is modified. Variables that are independent of the intervened variable will not change, so interventions allow us to estimate causal effects. Intervention is denoted by  $\text{do}(X = x)$ , the resulting probability distribution by  $P(Y \mid \text{do}(X = x))$ . It is important to note that  $P(Y \mid X = x)$  may reflect correlations and other confounding influences, so if we observe  $P(Y \mid X = x)$ , this is not necessarily the same as  $P(Y \mid \text{do}(X = x))$ . For example, the probability that you have a visitor given that you hear the doorbell ring is not the same if you get up and ring the doorbell yourself. In this case, the doorbell does not cause the visitor to appear.

Causality can be defined through the concept of interventions. We can say  $X$  causes  $Y$  if two different actions ( $x$  and  $x'$ ) result in two different distributions of  $Y$ :

$$\exists x, x' : P(Y \mid \text{do}(X = x)) \neq P(Y \mid \text{do}(X = x'))$$

## Adjustment sets

As mentioned in the previous section, adjustment sets are integral to understanding the causal metrics used in this work (see Sec.2.4). Often, we cannot directly intervene on a system. For example, we cannot change the temperature of a particular region to observe the impacts on other regions. Adjustment sets allow us to estimate what the probability distribution of a particular variable might be if we were to intervene, referred to as an interventional distribution, by using the available observational data. This estimation is performed by looking at a combination of observational distributions. Using adjustment sets is one strategy for identifying causal effects. The adjustment set itself is the set of variables we condition on to block all noncausal paths between the intervention variable and the outcome variable, which allows us to estimate the causal effect of  $X$  on  $Y$  without confounding influences. The adjustment formula in the discrete setting essentially averages the conditional probability  $P(Y \mid X = x, Z = z)$  over the distribution of each variable in the adjustment set  $Z$ . In other words, we marginalize out  $Z$  (thereby removing it from the joint distribution) to isolate the distribution  $P(Y \mid \text{do}(X = x))$ :

$$P(Y \mid \text{do}(X = x)) = \sum_z P(Y \mid X = x, Z = z) P(Z = z),$$

Given a causal graph, an identification strategy is a formula that estimates an interventional distribution from a combination of observational ones. There are multiple identification strategies including the backdoor and frontdoor adjustment (which use valid adjustment sets), instrumental variables, and randomized control trials, each of which apply in different settings.

### 2.3.2 Causal discovery

Causal discovery seeks to uncover the directional cause-and-effect relationships in observational data. In the causally sufficient case, these relationships are ideally expressed as DAGs, however, there are often multiple graphs that can represent the same sets of conditional independencies. Without further assumptions or domain knowledge, many algorithms can only go so far as finding a **CPDAG**, which represents a Markov Equivalence Class of DAGs. Methodologies for causal discovery fall into the following categories of structure learning algorithms:

**Constraint-based:** Use conditional independence tests to decide whether edges in a candidate graph explain dependencies observed in the data, or whether the edges are spurious and should therefore be pruned. Identifies up to the MEC. Representative algorithms include the Spirtes-Glymour-Scheine algorithm (SGS), Peter-Clark algorithm (PC), and Fast Causal Inference (FCI) [70].

**Score-based:** find the best causal graph that maximizes a score (often based on data likelihood). These methods often rely on strong assumptions, for example, incorporating background knowledge or knowing the parametric form of the underlying SCM. Identifies up to the MEC. One example is Greedy Equivalence Search (GES) [12].

**Restricted models:** By making assumptions about the types of functions that govern the underlying structural equations, or by making assumptions about the distributions of the exogenous variables, restricted models can recover DAGs from observational data. Algorithms in this class include RESIT for nonlinear additive noise [52] and LiNGAM for linear non-Gaussian models [68]. The strong assumptions, however, restrict the applicability of these methods.

**Interventional causal discovery:** In settings where there is available interventional (controlled intervention experiments) or heterogeneous data (data collected in different environments or experimental conditions), invariance-based techniques can identify causal parents of a target variable to output an interventional MEC (I-MEC). Examples include ICP [50], GIES [23], JCI [42].

#### Conditional independence tests

Constraint-based causal discovery algorithms rely on conditional independence tests. They check for conditional independence between pairs of variables such as  $X$  and  $Y$ , given a conditioning set  $Z$ . The set  $Z$  may be empty. The method tests the null hypothesis that variable  $X$  is independent of variable  $Y$  when conditioned on the set of variables  $Z$ :

$$H_0 : X \perp\!\!\!\perp Y | Z$$

The symbol  $\perp\!\!\!\perp$  means “independent of”. The symbol  $\not\perp\!\!\!\perp$  mean “not independent of”, or rather “dependent on”. A commonly used test is **partial correlation**, which computes the partial correlation between  $X$  and  $Y$  controlling for a single variable  $Z$ :

$$\rho_{XY|Z} = \frac{\rho_{XY} - \rho_{XZ}\rho_{YZ}}{\sqrt{1 - \rho_{XZ}^2} \sqrt{1 - \rho_{YZ}^2}}.$$

$\rho_{AB|Z}$  denotes the Partial correlation between variables  $A$  and  $B$  given  $Z$  [53]. The strength and direction of the linear correlation between two variables  $X$  and  $Y$  ranges from -1 to 1. A value of  $\rho_{XY|Z} = 0$  indicates no linear relationship between  $X$  and  $Y$  after removing the linear effects of  $Z$  through linear regression. In other words,  $X$  is conditionally independent of  $Y$  given  $Z$ . The partial correlation  $\rho_{AB|Z}$  may be computed given a set of variables  $\mathbf{Z}$  by recursively computing the partial correlations between  $X$  and  $Y$  given all subsets of variables.

The value  $\rho_{XY|Z}$  is then tested for significance, using tests such as Fisher’s  $z$ -transformation to obtain a  $p$ -value that indicates whether or not the observed correlation may have occurred due to coincidence.

Conditional independence tests vary in their distributional or functional assumptions. For example, partial correlation assumes linear relationships and a Gaussian joint distribution between the variables, making it unsuitable when those conditions fail. Alternative tests Chi-square for discrete data [49], and HSIC for nonparametric data [21].

### Causal discovery in time series

In this work we will focus on causal discovery in time series data, which is especially relevant for temporal modelling and forecasting in climate science. Notably, causal graphs for time series data exhibit a temporal ordering: cause must precede effect in time.

PCMCI is a widely applied causal discovery algorithm for time series [64, 14, 47, 58, 74]. In the climate science domain, it is used as benchmark for SAVAR [74] (described in Section 2.1.3, and applied in causal evaluation of DL emulators [58, 14, 47]. The Tigramite package provides an implementation of PCMCI.

### PCMCI Algorithm

The PCMCI algorithm (Peter and Clark Momentary Conditional Independence) [64, 61] is a constraint-based causal discovery method specifically designed for time series data. It consists of two stages: the  $PC_1$  step (pruning links) and the MCI step (verifying links).

**$PC_1$  step [61]:** This step removes spurious links between pairs of nodes that are independent. The algorithm iteratively runs conditional independence tests using larger conditioning sets, starting with an empty set. Instead of testing each possible combination of nodes, the algorithm tests candidate pairs in an optimized way. For each variable  $X_t^j$  (where  $j$  denotes the variable index, and  $t$  denotes the time index), we start with a candidate set of parents  $\hat{Pa}$  from previous time steps:

$$\hat{Pa}(X_t^j) = \{X_{t-\tau}^i : i \in \{1, 2, \dots, N\}, \tau \in \{1, 2, \dots, \tau_{\max}\}\},$$

where  $N$  is the number of nodes, and  $\tau_{\max}$  is the maximum number of time steps. Using a test such as Partial Correlation, the algorithm first tests unconditional independence ( $p = 0$ ) between  $X_t^j$  and each node in the candidate set of parents. If a candidate parent is independent of  $X_t^j$  (determined using a significance level  $\alpha_{PC}$ ), we prune it from the set  $\hat{Pa}$ . Each subsequent step increments  $p$  by 1, and performs a conditional independence test using a conditioning set with the  $p$  parents that have the highest marginal dependence values (obtained from the conditional independence test in the previous step). Each iteration removes candidate parents that are independent, until all that remains is a set of parents with significant dependence on  $X_t^j$ , conditional on all other parents in the set.

**MCI step [61]:** The  $PC_1$  step discovers up to a MEC, so the next step is necessary to reject false positive links that made it past the  $PC_1$  step, since time-series data is often highly inter-dependent. To test whether the link exists  $X_{t-\tau}^i \rightarrow X_t^j$ , the MCI step evaluates the following conditional dependence:

$$\text{MCI} : X_{t-\tau}^i \perp X_t^j \mid \hat{Pa}(X_t^j) \setminus \{X_{t-\tau}^i\}, \hat{Pa}(X_{t-\tau}^i)$$

Meaning we test whether  $X_{t-\tau}^i$  is independent on  $X_t^j$  when conditioned on both the parents of  $X_{t-\tau}^i$  and  $X_t^j$ , excluding  $X_{t-\tau}^i$  itself (since we are explicitly evaluating the link from that node). In other words, we test independence between two variables by conditioning on both sets of their parents (established in the  $PC_1$  step) to remove autocorrelation and confounding effects.

### Parameters

**Significance threshold  $\alpha_{PC}$ :** In conditional independence tests,  $\alpha_{PC}$  controls the trade-off between including weak but true links and avoiding spurious links. A too-low value of  $\alpha_{PC}$  might prune true links. A higher value of  $\alpha_{PC}$  may increase sensitivity but also runtime. The

$PC_1$  step essentially serves to reduce the dimensionality of conditions used in the MCI step, so  $\alpha_{PC} = 1$  would convert the MCI test to FullCI [64].

**Maximum time lag  $\tau_{max}$**  : Specifies the maximum number of time steps, or temporal window, over which one variable may causally influence another.

**Conditional independence:** The choice of test affects performance and errors depending on the type of data. Partial correlation works well for linear Gaussian systems, while kernel-based tests (GPDC) or mutual information (CMI) handle nonlinear relationships.

## Extensions

**PCMCI+** [61]. extends PCMCI to include [contemporaneous links](#) where cause and effect are observed instantaneously at the same time step. The algorithm also further optimizes the choice of conditioning sets, improving both reliability and efficiency of the algorithm.

**Multidata-PCMCI+** [60]. Whereas PCMCI infers causal structure from a single time series, Multidata-PCMCI+ analyzes multiple samples that share the same underlying process to estimate a single causal graph. This improves reliability when the individual samples do not have enough time steps for PCMCI to distinguish statistical significance from noise.

## Assumptions and Limitations

PCMCI assumes causal stationarity (relationships do not change over time), along with the standard causal discovery assumptions of causal sufficiency, faithfulness, and the Markov condition. The method can fail when sample sizes are too small, dependencies are very weak, or when causal sufficiency is violated (e.g., when there are unmeasured confounders). Additionally, PCMCI potentially scales poorly to high-dimensional systems due to the exponential growth of possible conditioning sets [64, 61, 58, 74, 8, 65].

One approach to addressing the limitations of causal discovery on high-dimensional data is to apply dimensionality reduction methods to the data before learning the causal graph. Simple methods for dimensionality reduction include spatial averaging and subsampling, but they are limited in their ability to capture complex dependencies. More sophisticated approaches include  $\delta$ -MAPS [58] and principal component analysis (PCA) followed by Varimax rotation [28, 74]. One challenge is learning disentangled representations, which means ensuring that the latent variables identified through the dimensionality reduction techniques each correspond to a distinct process, rather than a mixture of signals. However, the semantic interpretation of what the dim-reduced variables actually refer to is unclear. The question is whether the reduced signal is representative of patterns we are actually trying to learn.

### 2.3.3 Causal Representation Learning

Having discussed causal discovery methods that uncover causal structures directly between variables, this section introduces causal representation learning ([CRL](#)), a method that focuses on uncovering high-level latent factors within a system, and inferring their causal relationships. We further introduce CRL methods applied in this work.

#### Latent variables

So far, we have discussed causal discovery methods that identify relationships between variables that are present in the data. However, in climate science, data is often very high-dimensional. Causal discovery applied directly at the observational level could be computationally intractable, limited in interpretability, and could potentially obscure more abstract underlying factors that drive groups of variables. The NorESM climate model [67], for example, models variables on a grid at a resolution on the order of  $10^4$  to  $10^5$  pixels. For the context of this work, we are only looking at relationships within one particular type of measurement, for example, sea-level pressure, but the complexity of the problem increases dramatically when we want to consider interactions across several types of measured variables as well (e.g.,

temperature, precipitation, etc.).

In contrast to reducing the dimensions and applying causal discovery in two separate steps, Causal Representation Learning (CRL) is an approach that learns a smaller set of latent variables from high dimensional data while simultaneously drawing causal links between them [65, 7, 8]. Many CRL methods leverage machine learning methods such as deep networks and generative models to learn these complex latent representations [66]. Examples for temporal data include: Lippe et al. [38], Yao et al. [81], Klindt et al. [33], Lachapelle et al. [35].

As discussed in the Climate Science background 2.1, real-world climate dynamics are often shaped by unobserved processes or patterns, which are high-level abstract factors that influence the observed data. The El Niño–Southern Oscillation (ENSO) is one such example of high-level natural phenomena that can affect multiple variables at an observational level [72]. Instead of modeling relationships between every observed variable directly, it would be ideal if we could draw connections between these more abstract processes to gain a clearer understanding of the “big picture” dynamics in the climate system. Investigating whether CRL methods could learn suitable proxies for important climate patterns remains an open research challenge [8].

### Causal Discovery with Single-Parent Decoding

Causal Discovery with Single-Parent Decoding (CDSD) Brouillard et al. [8] is a causal representation learning method that is central to the PICABU model we use in this work. In contrast to methods like Varimax-PCMCi [74] which combine dimensionality reduction with causal discovery in two separate stages, Causal Discovery with Single-Parent Decoding (CDSD) is a CRL method that recovers disentangled latents from time series data and learns their temporal causal graph simultaneously via score-based structure learning (see Fig. 2.4). Notably, to demonstrate the applicability of CDSD for high-dimensional data science, Brouillard et al. [8] evaluate their method on a climate science task. The latent variables learned using CDSD end up being regional clusters, as variables within large regions are found to behave in a coherent way [74, 25]. These are not climate indices such as ENSO, however the authors find that the clusters correspond to established climatological regions [8]. For the data, assume a two-dimensional time series of  $d_X$  “observed” variables  $X$  at  $T$  time steps. In a climate science context, these variables might represent climate measurements or simulations (such as precipitation, temperature, etc.) at grid locations on Earth. The latent variables  $Z$  represent the unknown regional climate trends that govern the behaviour of the observed variables  $X$ . The method segments the observed variables, and these aggregated groups make up the latent variables. CDSD learns two graphs: 1) A causal graph  $G$  between the latent variables. In this graph, the past values influence the present across multiple discrete time steps, up to a maximum time lag  $\tau$ . 2) A bipartite matrix  $W$  that maps each observation  $X$  to a latent variable  $Z$ . In a climate science context, this would enable us to identify which observational variables on the map are governed by which latent variables. The graphs are learned simultaneously using continuous optimization.

**Assumptions.** CDSD assumes that there are no instantaneous connections that exist within the same time step. CDSD also allows for nonlinearity in the relationships (functions) between latents as well as the mapping from latents to observations. A key component of CDSD is a strong assumption that it makes, called the Single-Parent Decoding assumption. The method assumes that each observation  $X$  maps to at most one latent variable  $Z$ . By restricting each observation to have only one parent in the latent space, this assumption addresses the challenge of disentangling latent representation. The single-parent assumption ensures that latent variables can be uniquely recovered from the observed data, referred to as identifiability.

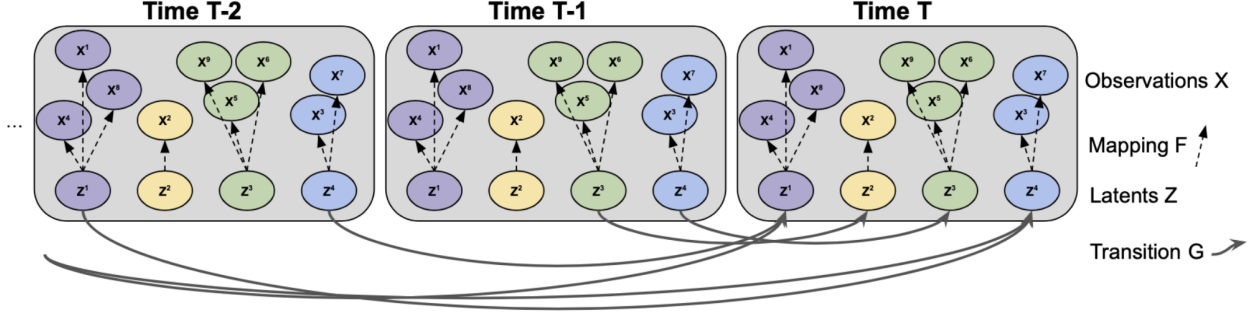


Figure 2.3: The variables  $z$  are latent and  $x$  are observed. The matrix  $F$  (referred to as  $W$  in this work, represented by dashed arrows) maps the latents to the observed variables, and the matrix  $G$  (full arrows) represents the time-lagged causal graph between latents. Only a subset of the full graph  $G$  is depicted to simplify the illustration. Taken from Hickman et al. [25] with permission.

**Penalties and Constraints.** CDSD enforces two key constraints to ensure identifiability and interpretability of the latent representations.

**Sparsity penalty:** To keep only important connections and reduce complexity, CDSD penalizes dense or excess connections on the latent causal graph  $G$ . Omitting the sparsity penalty removes the causal structure from the latent space, reducing the model to a standard VAE.

**Orthogonality constraint:** To ensure that the latents remain disentangled, the mapping  $W$  must satisfy non-negative and orthogonality constraints.

$$h(W) := W^T W - I_{d_z}$$

Where  $W \in \mathbb{R}^{d_x \times d_z}$  is the matrix mapping  $d_z$  latent variables to  $d_x$  observations, and  $I_{d_z}$  is the identity matrix of dimension  $d_x \times d_z$ . The orthogonality constraint enforces the Single-Parent Decoding assumption.

**CDSD Framework.** The CDSD model is built on a variational autoencoder (VAE) that jointly learns latent variables and their causal structure under a single-parent decoding assumption. Key components of the framework are:

**Encoder:** Maps high-dimensional observations  $x_t$  to a low-dimensional latent representation  $z_t$ :

$$q(z_t | x_t)$$

**Transition model:** Generates a future latent at the next time step by using the learned causal graph between latents  $G$ .

$$p(z^t | z^{<t}) := \prod_{j=1}^{d_z} p(z_j^t | z^{<t}),$$

Where  $z^{<t}$  refers to the variables that precede  $z^t$  up to a maximum lag  $\tau$ . Each conditional  $p(z^t | z^{<t})$  is parameterized by a nonlinear function between  $z^t$  and a transition function that takes as input  $G$  and the parents  $z^{<t}$ .

**Decoder:** Uses the single-parent decoding assumption to reconstruct each observed variable  $x_t^i$  from a single parent latent, which was generated using the transition model. The latent space is normally distributed.

$$p(x^t | z^t) := \prod_{j=1}^{d_x} p(x_j^t | z_{\text{pa}_j^F}^t).$$

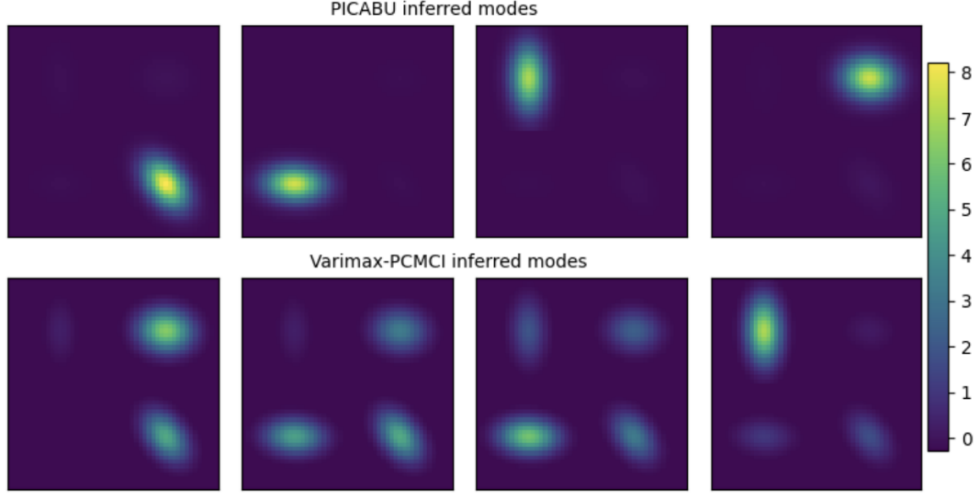


Figure 2.4: Example of how PICABU disentangles latents better than Varimax-PCA followed by PCMC1. Trained on SAVAR data with four latent variables with medium easy difficulty corresponding to link probabilities (see 3.2.1, PICABU can identify each latent as distinct from the others. Taken from Hickman et al. [25] with permission.

**Objective:** The training objective is to maximize the evidence lower bound (ELBO) for  $p(x^{\leq T})$ . The ELBO balances reconstruction accuracy with regularization of the latent space, and the loss is augmented with terms that enforce the sparsity penalty and orthogonality constraint.

### PICABU model

PICABU (Physics-Informed Causal Approach with Bayesian Uncertainty) [25] is an extension of CDSD that works both as a causal emulator and as a CRL method. This framework is specifically developed for climate applications, with a focus on interpretability and physical consistency in climate emulation. PICABU differs from CDSD in a few ways:

**Sparsity constraint:** PICABU constrains sparsity instead of penalizing by using domain knowledge to enforce an expected number of links in the latent graph  $G$ . Hickman et al. [25] report that this improves model convergence.

**Physics-informed loss** The training objective includes additional loss terms to ensure that the learned dynamics respect known physical patterns. The spectral loss ensures that the model captures the correct spatial and temporal power spectral densities of the data, and the CRPS loss is a probabilistic measure of accuracy for the next-timestep prediction.

**Emulation:** The model implements a Bayesian filter for stable autoregressive rollouts, however, for the purpose of this work, we will largely focus on its CRL capabilities.

### Limitations.

**Single parent decoding:** Although this restriction enables learning disentangled latents, the assumption may oversimplify systems in which variables are influenced by multiple factors.

**Hyperparameter sensitivity:** PICABU is sensitive to the weighing of terms in the loss. Reliance on domain knowledge to enforce the sparsity constraint also limits applicability in settings where the expected number of links in the latent causal graph are not known.

**Structural learning only, no Parameterization:** PICABU does not learn the weights of links, only binary causal graphs. This limits the analysis of the strength or parameters of causal connections.

## 2.4 Causal graph comparison

The overall goal of this work is to evaluate how well a machine learning model can learn the underlying causal connections of a system. To do this, we will use the above methods (dimensionality reduction with causal discovery, and causal representation learning) to infer a causal graph associated with the emulated system, and a second causal graph associated with the ML emulator prediction. We expect that the distance between the two graphs will reflect the emulator robustness and go beyond traditional performance metrics. In this section, we will highlight different distances between causal graphs.

Before diving into the metrics, an important assumption in this work needs to be clarified: We stated earlier that score-based and constraint-based methods learn graphs up to a MEC (see Section 2.3.1). Our methods are not exempt from this, without additional assumptions. This raises the question: how can we compare graphs by structure if we cannot guarantee that we are learning the correct orientations? In our case, there is only one graph per MEC because: (1) all the arrows go forward in time (cause must precede effect temporally), and (2) since we assume that there are no instantaneous connections (contemporaneous links), there are no cases in which edge orientation could go either way, which would otherwise be in a CPDAG. Because of the assumptions we make about causal sufficiency, the temporal direction of causality and the absence of instantaneous connections, the MEC discovered by our methods contain one DAG.

### 2.4.1 Structural Metrics

**Structural Hamming Distance (SHD [51])** is a straightforward comparison that counts the number of node pairs  $(i, j)$  that disagree on edge type between two graphs  $G$  and  $\mathcal{H}$  given by:

$$\text{SHD} : \mathbb{P} \times \mathbb{P} \rightarrow \mathbb{N}$$

$$(G, \mathcal{H}) \mapsto \#\{(i, j) \in V \times V \mid G \text{ and } \mathcal{H} \text{ do not have the same type of edge between } i \text{ and } j\}$$

Differences include edge additions, deletions, or inverted directions. While efficient to compute and easy to implement, SHD focuses solely on structural differences and does not sufficiently capture the causal implications of these differences. For example, a seemingly small structural change (small SHD) may have strong causal implications.

**Modified F1 Score [47]** balances precision (fraction of predicted links that are correct) and recall (fraction of true links that are detected). In climate applications, this metric is often modified to take into account the sign of the link (positive or negative influence), and to consider a link as correct if it is found within a specified temporal tolerance of the reference link. The F1 score ranges from 0 (no match) to 1 (perfect match) but, like SHD, focuses purely on link existence rather than causal implications.

### 2.4.2 Adjustment-Based Causal Metrics

**Structural Intervention Distance (SID [51])** addresses the limitations of structural metrics by counting pairs of variables for which intervention distributions  $P(Y \mid \text{do}(X = x))$  differ between the true and estimated graphs. In other words, SID counts the number of parent sets in the predicted graph that are not valid adjustment sets in the true graph. SID evaluates graphs by comparing their causal inference statements: two graphs have  $\text{SID} = 0$  if they yield identical intervention distributions for all variable pairs, even if they are structurally different. However, SID is asymmetric (meaning the SID score may differ for the same pair of graphs depending on which graph is being used as the reference graph, which qualifies it only as a pre-metric). Furthermore,  $\text{SID} = 0$  does not imply that the graphs are necessarily structurally

equal, only that they have the same intervention distributions.

**Adjustment Identification Distance (AID [24])** generalizes SID by incorporating different adjustment strategies (Parent-AID, Ancestor-AID, or OSet-AID (based on optimal adjustment sets)) to derive identification formulas for causal effects. AID compares graphs by checking whether identification formulas derived from one graph remain valid in another. This framework scales better than SID while providing a more nuanced evaluation of causal inference capacity. For DAGs, Parent-AID is equivalent to SID. A property of the AID pre-metrics is that if a DAG  $G_{true}$  is a subgraph of the DAG  $G_{guess}$ , the identification distance will be 0 [24]. Additionally,  $SHD = 0$  implies that  $AID = 0$ , however the other way around does not hold.

### 2.4.3 Other causal metrics

**Distance Average Causal Effect (DACE [58])** Average Causal Effect (ACE) measures how important a node is in acting as a causal driver of the variability in a system by assessing how perturbations flow through a causal network. Each node has a particular Average Causal Effect value, which is the mean of the effect of a particular node on every other node. The estimated Causal Effect of perturbations, assumed to be linear, corresponds to the normalized strength, or weights, of the causal dependencies.  $DACE = 0$  indicates complete agreement between graphs. This metric stands out among the others by considering the weights of links in causal graphs.

### 2.4.4 Separation-Based Metrics

**Separation Distance (SD [77])** measures disagreements in conditional independence statements between graphs rather than causal effects. SD applies different separation strategies to identify conditioning sets in a graph, and verifies whether those sets maintain their separating property in the reference set. Unlike intervention-based metrics, SD is invariant under Markov equivalence, meaning graphs with identical conditional independence structures are considered equivalent.

## 2.5 Related work

Existing literature [30, 14, 58, 47] in process-oriented evaluation of climate models have adopted a causal model evaluation (CME) framework proposed by Nowack et al. [47]. All methods share the underlying assumption that similarity in causal representations between a model and those derived from the reference data reflect fidelity to the internal dynamics of the reference [63]. First, dimensionality reduction is applied to both model projections and the reference data. Next, a causal discovery method is used to discover time-lagged causal relationships, which are then compared using graph comparison metrics. Finally, whether causal metrics do indeed capture faithful internal dynamics is then validated through various diagnostic methods.

One key distinction in these works from what we have been discussing so far is that the CME framework is applied to evaluate numerical climate models from the CMIP6 or CMIP5 sets, against reanalysis data. Reanalysis data interpolates incomplete observational data using numerical models [2]. Dimensionality reduction techniques vary across sources. Debeire et al. [14] and Nowack et al. [47] use Varimax-PCA 2.3.3 to uncover latent modes, whereas Ricard et al. [58] use the  $\delta$ -maps method to cluster grid cells that covary into geographic regions. The difference lies in that latent modes do not necessarily refer to geographically continuous regions, as assumed by Ricard et al. [58]. Karmouche et al. [31] instead apply the Climate Variability Diagnostic Package for large ensembles (CVDP-LE) [54] to compute time series indices of four major modes of internal climate variability from reanalysis data. Each index has a single signal, so the dimensionality is therefore appropriate for causal discovery. For causal discovery, all works use PCMCI [64], although Karmouche et al. [31] use PCMCI+ [61]. Karmouche et al. [31] and Debeire et al. [14] both use modified F1 scores introduced by [47] described in

2.4. Ricard et al. [58] use WWD and DACE described in 2.4 to quantify distance between models and their references. The causal metrics are then used to weigh models in multi-model projection tasks, demonstrating lower projection uncertainty in weighted ensembles compared to baseline [14, 58, 47].

Tibau et al. [74] provide a benchmark for causal discovery on SAVAR data at the mode level. The method applies Varimax-PCA for dimensionality reduction followed by PCMCI for causal discovery at the mode level. Fidelity of link detection is evaluated use precision and recall separately. Hickman et al. [25] (Accepted at NeurIPS 2026) introduce PICABU, a causal emulator with causal representation learning capabilities. The authors do not compare the learned causal graphs on emulation tasks, rather evaluating performance on other measures such as power spectra. However, they evaluate PICABU as a causal representation learning method for recovering the mode-level ground truth causal graph on data generated by SAVAR using F1-score.

A persistent challenge across existing efforts is the difficulty of effectively comparing causal graphs once inferred. Structurally similar graphs can have very different causal implications [51], so while the commonly used F1 score metric is easy to implement, it may fail to capture causally important differences between causal graphs. Debeire et al. [14] acknowledge the limitation of using F1 scores, and suggest that metrics that consider causal effects could be better. In their review on causal inference frameworks for climate science, Runge et al. [63] similarly suggest that causal comparison metrics should be used for comparing causal networks. Promising alternatives include (1) Adjustment Intervention Distance (AID) [24], which evaluates differences in causal effects across interventions, and (2) distance average causal effect [58], which compares the weighted causal importance of nodes between graphs.

Our approach differs from prior work in several key aspects. First, we focus on evaluating DL emulators against their reference ESMs, rather than comparing ESMs to reanalysis data. To validate our methods in a controlled environment where we have access to ground truth causal graphs, we use SAVAR [74] to generate our reference data. We compare two causal inference methods: (1) the first combines dimension reduction via weighted averaging with causal discovery using Multidata-PCMCI+ [60] and (2) the second method employs causal representation learning through PICABU [25], which addresses the limitations of (1) in disentangling latent variables as discussed in Section 2.3.2. For causal graph comparison, we apply the Parent-AID metric [24] described in 2.4. Although investigating how well causal functions (link weights) are captured is essential, it remains beyond the scope of this work and is left for future research.

To validate the utility of our using causal metrics in a CME framework, we assess whether they are predictive of “good” model performance, benchmarked by RMSE and analysis of power spectral density [56]. We then leverage SAVAR’s capacity to simulate interventions on its causal graph to test whether causal metrics computed pre-intervention better predict emulator robustness compared to structure-based metrics such as F1 score.

It is important to acknowledge the major limitations and assumptions of this work. (1) This approach is sensitive to the causal discovery method used. Challenges can range from tuning methods to assumptions about linearities and distributions in the data, sensitivity to dimensionality reduction methods, difficulties in disentangling latents, addressing stationarity and high-interdependence between variables, unknown confounders, and performance capacity of the model to learn the “correct” underlying graph. (2) PICABU specifically makes a strong single parent assumption which, although valid in the SAVAR setting, may not necessarily hold in real-world climate systems. (3) Finally, a strong assumption of the CME framework is that the Earth system has causal relationships between latent variables that are not directly observable, but that govern dynamics of observed variables. This assumption grounds the premise that comparison of causal graphs can be used to assess the similarity between two climate models [47].

# Chapter 3

## Methodology and Implementation

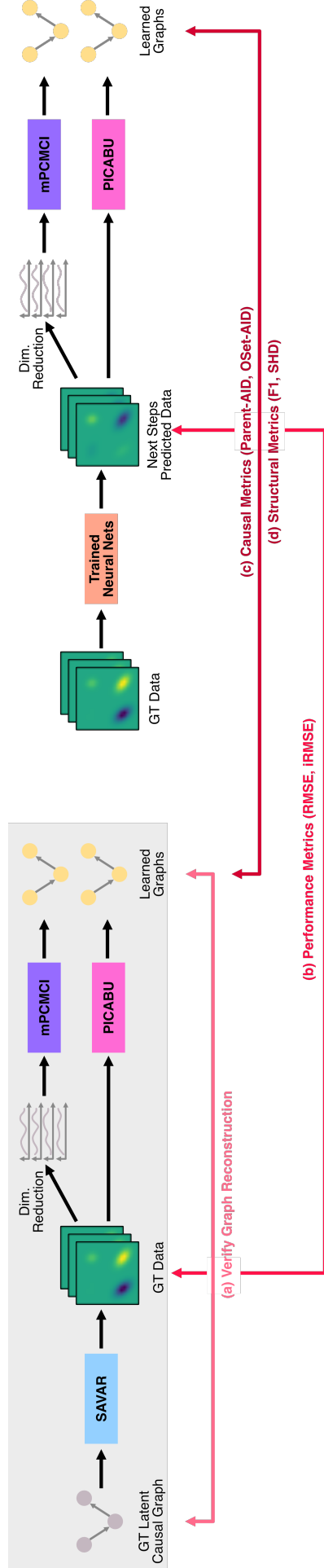
The goal of this work is to determine whether causal metrics are more suitable than structure-based graph comparison metrics for the Causal Model Evaluation (CME) framework proposed here. The CME framework compares the causal relationships implicitly learned by an emulator to the ones inherent in the training data. We evaluate whether causal metrics (as opposed to structural metrics) could be suitable indicators of good emulator performance by examining how strongly they correlate with other metrics such as [RMSE](#), spectral fidelity, and intervention RMSE.

The following is a summary of our methods; further details are provided in the methodological and implementation sections below. We train a set of next-step prediction emulators (neural networks), each designed to capture different aspects of the data. These models are trained on data generated by [SAVAR](#), a simplified climate model that simulates spatial time-series data with known causal relationships [74]. We generate projections (predictions for future timesteps) with the chosen emulators. From these projections, we learn causal graphs over the latent modes that govern the high-dimensional (grid-level) variables. We use two methods for learning the causal graph underlying SAVAR, which generated data with distinct, non-overlapping modes: (1) We perform dimensionality reduction on the data by taking the weighted average of each grid-level variable within a mode. We then apply [PCMCI+](#) for multivariate time series [60, 62] to learn the causal graph over multiple samples with 20 time steps. In this work, we refer to the combination of dimensionality reduction followed by Multivariate-PCMCI+ as “[Avg-PCMCI](#)”. (2) We use [PICABU](#) to learn the latent causal graph by training on multiple samples, each containing sequences of five previous time steps and the corresponding next step. We apply these two methods to learn the latent causal graphs from both the reference data, and the emulator projections. We then compare the two causal graphs using [Parent-AID](#) [24], a causal graph comparison metric, as well as [F1-score](#) [47] and [SHD](#) [24], which are structural graph comparison metrics. The graph metrics are compared against three model performance metrics: (1) Root Mean Square Error (RMSE) of the next-step prediction; (2) Least Squares Distance ([LSD](#)) between the [FFT](#) coefficients of 20-step autoregressive rollouts and the ground truth; and (3) RMSE between the model output with intervened input data and the corresponding ground truth next step from the intervened data. In total, we obtain six graph metrics for each trained model (Parent-AID, F1 score, and SHD on graphs obtained by both PICABU and Avg-PCMCI) and three model performance scores: RMSE, LSD, and Intervention RMSE.

### 3.1 High-level Overview of the Causal Model Evaluation Framework

The following section as well as [Fig. 3.1](#) are intended solely as a high-level overview to orient the reader; further methodological and implementation details are provided in subsequent sections.

**Step 1: Learn a latent causal graph on spatiotemporal data.** We begin with univariate spatial time series data generated by a model. Because this study requires generating ground truth next steps for intervened data (as detailed in [Step 7](#)), we cannot use observational data. Using a causal representation learning (CRL) method, or a heuristic approach to CRL featuring



**Figure 3.1: Method Overview:** Left side, grey background: We use SAVAR, a simplified climate model, to generate 2D time series data from a known latent causal graph. We use two methods to obtain inferred causal graphs: dimensionality reduction in combination with Multidata-PCMCi+ (mPCMCi), and PICABU. We compare the learned graph to the ground truth graph (a) to verify accurate reconstruction for each method. Right side: We use the SAVAR-generated data to train several neural networks. These are in turn used to generate predicted next steps in the 2D timeseries. As before, we apply dimensionality reduction + mPCMCi and PICABU to obtain inferred causal graphs. The main focus of this work is to investigate if the performance metrics (b) correlate with causal metrics (c) or structural metrics (d), i.e. if we can use causal or structural graph comparison methods to infer a model’s predictive performance.

dimensionality reduction and causal discovery, we learn a causal graph over [latent variables](#) that captures the underlying system dynamics in the data.

**Step 2: Train a neural network for emulation.** We then train a neural network emulator on the high-dimensional time series from Step 1. The emulator is a next-step prediction model, which takes as input the previous five time steps and outputs the next step.

**Step 3: Learn the causal graph on trained emulators.** We then apply the same CRL method from Step 1 to learn a causal graph over latents from emulator-generated projections. Projections are either 1-step or multi-step autoregressive rollouts, depending on whether a CRL or heuristic CRL method is used respectively.

**Step 4: Compare graphs using causal and structural metrics.** We compare the graphs obtained in Steps 1 and 3 using causal and structural graph comparison metrics. Graphs learned using CRL methods are first permuted to ensure alignment between latents, enabling meaningful comparison between the learned graph and the reference.

**Step 5: Evaluate emulators using performance metrics.** We evaluate emulator performance by computing the RMSE between the predicted and true values for the next time step, as well as Least Squares Distance (LSD) between emulator and reference Fast Fourier Transform (FFT) coefficients obtained from autoregressive rollouts.

**Step 6: Evaluate emulator performance RMSE on intervened data.** We perform a soft intervention on the input to the previously trained emulator from Step 3, modifying the high-dimensional values corresponding to a single mode at a specific time step within the input sequence. The ground-truth next step for the intervened data is generated using the same model that generated the original dataset in Step 1. We then evaluate the difference between the emulator’s prediction and this target by computing the RMSE.

**Step 7: Compare graph metrics to emulator performance metrics.** We assess correlation between the causal and structural graph comparison metrics (step 4) against the emulator performance metrics (obtained from steps 5 and 6).

## 3.2 Methods

This section outlines the methodological framework used to evaluate emulators through causal graph comparison. We describe the data, models, and evaluation metrics employed. Detailed implementation specifics are provided in the following section.

### 3.2.1 Dataset generation

We use the Spatially Aggregated Vector-AutoRegressive (SAVAR) model to generate simulated data for this study [74]. For a further description of SAVAR, please see Section 2.1.3. This section explains the data referred to in Step 1 in the high level overview.

To recap, SAVAR is an [SCM](#) that generates two-dimensional time series using a known causal graph. An example sequence of the gridded SAVAR data is illustrated on the first row in Fig. 3.3. The ground truth causal graph represents time-lagged relationships between non-overlapping latent modes that govern the behaviour of the high-dimensional data at the grid level. Access to this ground truth graph is why we have chosen to use SAVAR, as it enables us to validate the methods we use to learn the causal graph. If we cannot trust that our methods learn latent causal graphs accurately, downstream analysis becomes irrelevant. The predicted next step in the data generation pipeline can be expressed as the sum over time lags:

$$X_{t+1} = \sum_{\tau=0}^{\tau_{\max}-1} W^\dagger G_\tau W X_{t-\tau} + \varepsilon_{t+1}, \quad (3.1)$$

$$\varepsilon_{t+1} \sim \mathcal{N}(W\mu_x, \Sigma_x) \quad (3.2)$$

Where:

$X \in \mathbb{R}^{\tau_{\max} \times n \times d_x}$  is the data tensor, where  $\tau_{\max}$  is the maximum lag,  $n$  is the number of features or types of measurements (in our case  $n = 1$ ), and  $d_x$  is the number of observed variables.  $W \in \mathbb{R}^{d_x \times d_z}$  denotes the orthogonal mapping matrix from latent variables  $z$  to the high-dimensional variables  $x$ , with  $W^\dagger$  representing its pseudoinverse.

$G_\tau \in \mathbb{R}^{d_z \times d_z}$ ,  $\tau = 0, \dots, \tau_{\max} - 1$  is the time-lagged adjacency matrix that represents the relationships between modes.

$\epsilon_\tau$  is independently sampled Gaussian noise.

The generative process of SAVAR is thus linear, with added noise. The causal graph is a temporal DAG with directed arrows between  $x_{t-\tau}^i \rightarrow x_t^j$  for  $\tau > 0$  and nonzero  $(W^\dagger G_\tau W)_{ij}$  [74]. In other words, links cannot go backwards in time, and there are no instantaneous connections, meaning that causal links always occur with a time lag. Variables at individual grid points within the same mode region exhibit coordinated changes that reflect emergent behaviour. The causal graph captures the time-lagged causal effects between these modes, which drive grid-level dynamics, rather than representing causal interactions between individual variables.

Excluding noise terms for brevity, the ground truth latent equation that generates the causal graph in the medium easy setting in Fig. 3.2 is:

$$\begin{aligned} X_0(t) &= 0.5X_0(t-2) + 0.48X_2(t-4) \\ X_1(t) &= 0.17X_1(t-2) + 0.28X_0(t-4) \\ X_2(t) &= 0.29X_2(t-3) + 0.36X_0(t-5) \\ X_3(t) &= 0.17X_3(t-3) + 0.68X_2(t-5) \end{aligned}$$

where each variable  $X_i$  represents the mode that governs a particular subgrid, and the time step is represented in brackets. The coefficients represent the weight or influence of each mode on  $X_i(t)$ .

There are 4 difficulty settings, which correspond to link densities in the ground truth causal graph: easy, medium-easy, medium-hard, and hard. The difficulty setting modulates the probability that a particular mode influences another in the future.

| Difficulty | Modes | Probability | General Equation                                 |
|------------|-------|-------------|--------------------------------------------------|
| Easy       | 4     | 0           | 0                                                |
| Med Easy   | 4     | 0.333       | $1/(N-1)$                                        |
|            | 16    | 0.067       |                                                  |
| Med Hard   | 4     | 0.667       | $2/(N-1)$                                        |
|            | 16    | 0.133       |                                                  |
| Hard       | 4     | 1           | $1/(a)$ where $a = 1$ for $N \leq 4$ ,<br>else 2 |
|            | 16    | 0.5         |                                                  |

Table 3.1: Probability of there being a link  $x_{t-\tau}^i \rightarrow x_t^j$  for  $\tau > 0$  in the ground truth graph generated by SAVAR, ordered by difficulty and number of modes.

In the easy setting, where the probability of a mode connecting to any other mode is 0, the causal graph defaults to the baseline in which each mode has one link to itself at a previous time, referred to as a “lagged auto-correlation”. Where there are fewer than 5 nodes, we ensure that every node is connected to every other node. For 5 or greater nodes, this would be infeasible, so the link probability is  $1/2$ . The difficulty settings thus affect the link density, or interconnectedness of nodes. The expected number of links in the true causal graph is  $M = N + N \times (N-1) \times p$ , where  $N$  is the number of modes and  $p$  is the link probability.

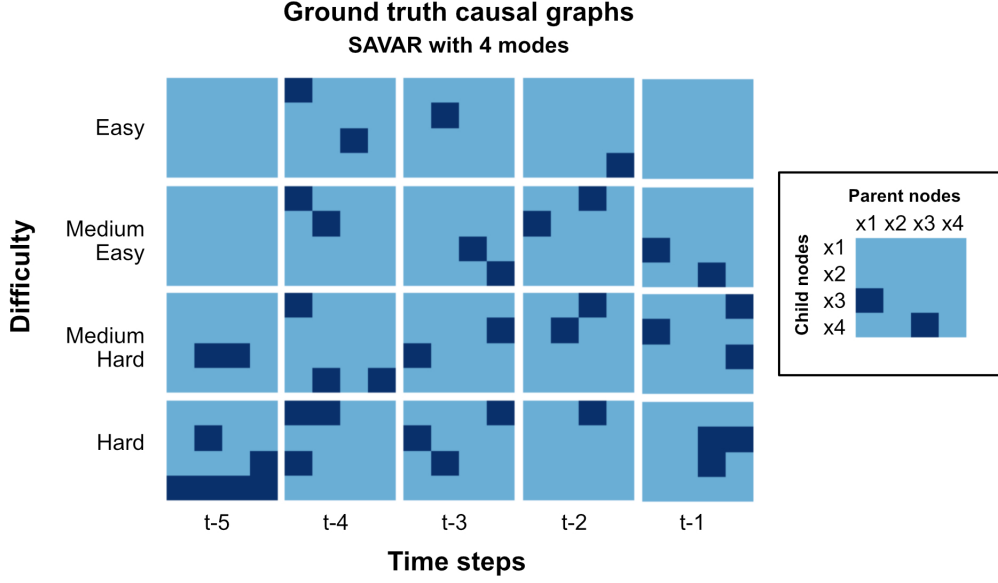


Figure 3.2: **Ground truth causal graphs generated by SAVAR for 4 modes with varying difficulties.** The causal graphs are represented as temporal adjacency matrices, where columns represent consecutive time steps from  $t-5$  to  $t-1$ , approaching  $t$ . In each matrix, the columns represent parent nodes, while rows indicate child nodes. Dark blue squares denote the presence of a causal link between nodes, while light blue squares indicate the absence of a link. For example, shown in the legend box on the right: there is a causal link from node  $x1$  to node  $x3$ , and another from node  $x3$  to node  $x4$ .

### 3.2.2 Training the Emulators

This section refers to Step 2 in the high level overview. For the purpose of this work, we choose to use simple models to emulate SAVAR, instead of complex DL-based atmospheric emulators which would be difficult to implement and computationally costly to train. We use four different ML emulators, each of which capture different spatio-temporal characteristics of the data through their different architectures - a simple Multi-Layer Perceptron (MLP) that captures non-linearities [59], a Long Short-Term Memory Network (LSTM) that captures sequential data [26], a Convolutional Neural Network (CNN) that handles spatial information [36], and a Variational Autoencoder (VAE) that specializes in discovering latent representations [32], the latter adapted from the PICABU architecture [25]. Each model is a next-step prediction model, trained on an input of 5 sequential time steps to predict the next step. Figure 3.3 illustrates a sample input-output sequence for each model, showing the 5-step input sequence alongside the predicted next step.

We do not expect these ML models to perform optimally; rather, the purpose is to understand how we can evaluate the emulators' capabilities. Consequently, we carry out only minimal hyperparameter tuning to prevent overfitting. In cases of slight overfitting, we use the best model before validation loss increases.

### 3.2.3 Learning the latent causal graph

This section refers to step 3 in the high level overview. Since the causal graph generated by SAVAR exists at the mode level, we need to learn the graph over latent variables, not directly on the high-dimensional variables. As discussed in Section 2.3.2, causal discovery is inappropriate for this task.

We use two methods to learn the causal graph over latents to compare methods. In both methods we assume stationarity in the data, meaning that the causal relationship of any given pair of latent variables does not change across time. The methods also make the following assumptions about the data (all discussed in Section 2.3.1): faithfulness, causal sufficiency,

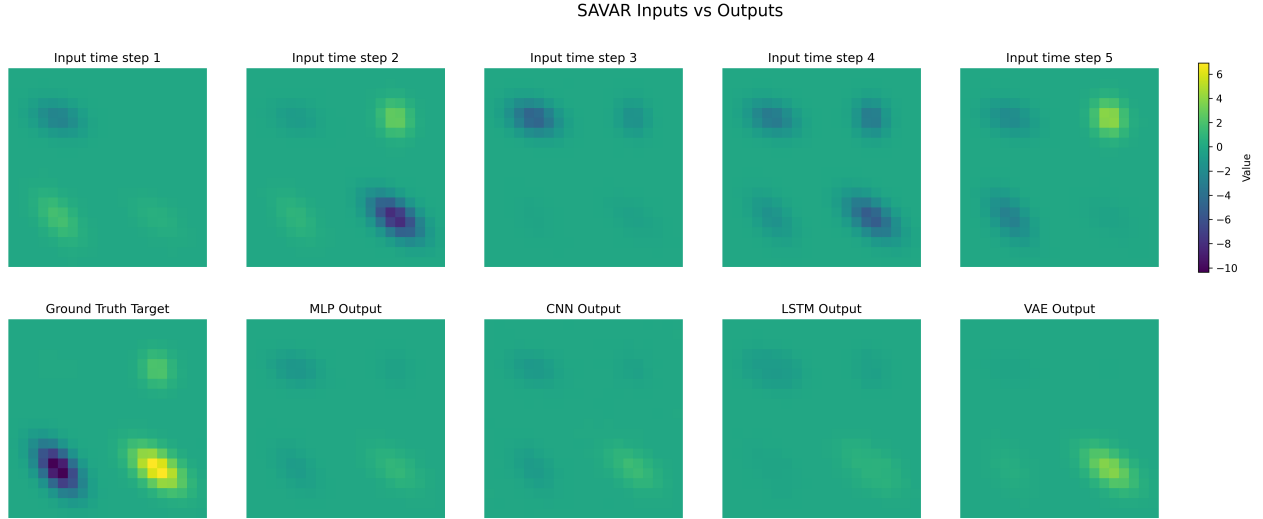


Figure 3.3: **Example Rollout of SAVAR Data.** Top row: sequence of 5 time steps generated by SAVAR for 4 modes on medium-hard difficulty. Bottom row: 6th time step from the dataset (left) and predicted by different emulators (MLP, CNN, LSTM, and VAE respectively).

causal Markov condition, no contemporaneous effects.

### Dimensionality reduction with Multidata-PCMCI+

We choose Multidata-PCMCI+ [60] as a method to recover causal graphs from the data following dimensionality reduction because PCMCI has been applied in a wide range of Causal Model Evaluation frameworks for climate-specific tasks including [47, 30, 14, 58, 74].

**Step 1: Generate time series.** First, we generate time series consisting of 20-step autoregressive rollouts produced by the emulators. This process is repeated over 1000 samples, chosen to match the number of samples obtained by partitioning the data into 20-step sequences using sliding windows. This approach ensures a fair comparison between the data and the emulator projections. We experimented with shorter sequences (ideally to learn a mapping between 5 input time steps and the next step), but this approach was unsuccessful as PCMCI+ needs a long-enough time signal to learn accurate relationships.

**Step 2: Apply dimensionality reduction.** As the ground truth causal graph for SAVAR is defined at the mode level, while PCMCI+ attempts to infer causal graphs among all input variables, we must first apply dimensionality reduction to the data as a preliminary step. Additionally, learning the causal graph at the grid-level would be both infeasible and undesirable; PCMCI+ is not well suited for high-dimensional spaces, and interpreting causal graphs with over 400 variables is impractical. To address this, we reduce dimensions by aggregating local spatial regions that correspond to a particular mode (e.g.,  $10 \times 10$  grid cells) into single representative variables. For each spatial quadrant, we compute a weighted average of the variables using Gaussian weights centered within the quadrant. This approach emphasizes central values, where variability is greatest, while down-weighting background zeros and peripheral noise. For data with four modes, for example, this produces a 20-step signal with four variables.

We initially attempted to apply Principal Component Analysis (PCA) [76] over the entire map, but this failed to disentangle the latent modes meaningfully. We also experimented with spatial subsampling by taking the central pixel of each quadrant, but this method is less appropriate from a climate science perspective. We did not use PCA followed by Varimax rotation [28] as in the SAVAR paper [74] or the benchmark method compared to PICABU [25] because the method does not effectively disentangle latent variables (see Fig. 2.4). This results

in latent variables that do not correspond to distinct modes, causing PCMCI to learn mixed signals. Since our goal is to compare causal graphs rather than improve methods for learning latents, we use a simpler method that, while less realistic, provides clearer mode separation for meaningful comparison.

**Step 3: Apply Multidata-PCMCI+.** We use PCMCI+ due to its increased efficiency, however without allowing for [contemporaneous links](#) (0-lag). For a full description of the functionality of PCMCI+ and its Multidata setting, see [2.3.2](#). The PC step in PCMCI can use different independence tests; we use partial correlation (ParCorr) in this study due to its computational efficiency. ParCorr assumes that the joint distribution between variables is Gaussian, and that the associations between variables is linear. This aligns well with the characteristics of the SAVAR-generated data, which follows a linear Gaussian structure.

**Step 4: Filter learned graph.** The Multidata-PCMCI+ method learns more links than are present in the ground truth graph. Using the expected number of links, we filter the learned graph by retaining only the strongest connections based on their weights, derived from their associated p-values that represent significance of the causal dependency.

## PICABU

We choose to use PICABU as our CRL method because it disentangles latent variables well [\[25\]](#). As previously discussed, the causal graph we are learning on SAVAR data is over latent variables, so CRL is an appropriate method for this setting. CSDS could have been used, but PICABU constrains sparsity directly rather than penalizing it, allowing us to set sparsity equal to the expected number of edges in the true graph. A limitation of PICABU is that it does not learn edge weights, so it does not estimate parameters for the causal connections, which constrains our choices of causal metrics, ruling out using metrics such as [DACE](#). For a full description of PICABU, see [Section 2.3.3](#).

**Step 1: Learn the reference graph.** PICABU learns the causal graph by mapping from 5 input time steps to the next step. This procedure is applied to learn the reference graph on the SAVAR data.

**Step 2: Learn the emulator graph.** To learn the graph implicit in emulator projections, we keep the 5 input time steps but replace the next step with the next step prediction from the emulator.

**Step 3: Permute the learned matrices.** PICABU learns graphs up to a permutation, meaning that the modes do not necessarily align with the reference graph. In other words, there is no guarantee that  $X_1$  in the learned matrix refers to  $X_1$  in the reference graph. We thus permute the learned matrix to align the inferred modes with the ground truth modes. The function (which is part of the PICABU Python package) does so by minimizing the squared Euclidean distances between mode coordinates, then applies that permutation to reorder the adjacency matrices across all lags.

### 3.2.4 Graph comparison metrics

We conduct two types of graph comparisons. First, we compare the graph inferred from SAVAR data to its corresponding ground truth graph ((a) in [Fig.3.1](#)). Second, we compare the graph inferred from the emulator to the graph inferred from SAVAR data ((c,d) in [Fig.3.1](#)). For the latter, comparisons are restricted within each graph-learning method: graphs learned by PICABU are compared only to other PICABU-inferred graphs, and likewise for Avg-PCMCI. This section refers to step 4 in the high level overview.

**Step 1: Binarize learned graphs.** The Avg-PCMCI method learns weighted adjacency matrices, so we first convert them to binary. PICABU learns binary matrices.

**Step 2: Flatten learned graphs.** The causal metric Parent-AID works on two-dimensional matrices. We therefore flatten the learned graphs retaining their temporal structure using methods described in [Section 2.3.1](#)

**Step 3: Apply structural metrics.** We apply SHD (See 2.4.1, and F1 score to compare graphs. For simplicity, we use the standard F1 score, not the modified score introduced by Nowack et al. [47]. Expressed in terms of true positives (TP), false positives (FP), and false negatives (FN):

$$\text{F1 score} = \frac{\text{TP}}{\text{TP} + \frac{1}{2}(\text{FP} + \text{FN})}$$

**Step 4: Apply causal metrics.** We compare graphs using Parent-AID and OSet-AID [24].

### 3.2.5 Emulator performance metrics

#### Experiment 1: RMSE

In a climate science context, the Earth might be partitioned into a grid based on latitude and longitude. Here, cell areas vary, so RMSE would need to be adjusted to take into consideration the proportional contribution of each cell, as discussed in Section 2.1.5. Since SAVAR data is represented on an evenly spaced grid, such adjustment is unnecessary. Therefore, we compute the standard Root Mean Squared Error (RMSE) as follows:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}$$

Where  $\hat{y}$  represents the predicted next-step value. This experiment refers to step 5 in the high level overview.

#### Experiment 2: LSD

LSD is a common metric used to assess the physical fidelity of emulator by evaluating how well it captures different spectral frequencies in the data [25]. See section 2.1.5 for more discussion. This section refers to step 5 in the high level overview.

**Step 1: Obtain 20 sequential time steps from SAVAR data.** We partition the data generated by SAVAR into samples of 20 time steps.

**Step 2: Run autoregressive rollouts.** Run simple autoregressive rollouts of 20 time steps on all models. At inference, we feed 5 sequential inputs into our model and the model generates a prediction for the next step. To continue the autoregressive rollout, we remove the first input from the original sequence and include the model’s output as new last element. This is fed into the model again and the process is repeated until we have 20 predicted time steps.

**Step 3: Apply dimensionality reduction on rollouts.** Apply dimensionality reduction per mode to obtain a weighted mean using Gaussian weights, following the same method described for Avg-PCMCI. The result is one signal of 20 time-steps per mode. This is done on both the SAVAR data from Step 1, and the autoregressive projection from Step 2.

**Step 4: Apply Fast Fourier Transform.** Apply Fast Fourier Transform (FFT) to the generated timeseries, as well as the ground truth SAVAR data. We average the frequency coefficients of the FFT for each mode across all samples.

**Step 5: Calculate Least Squares Difference (LSD).** Using averaged coefficients, we calculate the least squares difference between the absolute value coefficients for rollouts and ground truth. This gives us an LSD value for each mode. We report the mean over all modes.

#### Experiment 3: Intervention RMSE

This section refers to step 6 in the high level overview. This performance evaluation method can demonstrate how skillfully climate models represent the dynamics of the climate system, by showing how projections would change if past measurements were different. Similar to pacemaker experiments, a counterfactual method that prescribes specific values to a climate

variable [15], the question this experiment addresses is “Do the emulator’s predictions change under intervention in the same way as SAVAR’s predictions under (the same) intervention?”

**Step 1: Intervene on SAVAR data.** At inference time, we modify the inputs to the model. The inputs are the 5 time steps. We randomly select one mode at one time step to intervene. For a 4-mode dataset, for example, we may choose to intervene on mode 2 at time step  $t-3$ . We then perform a soft intervention by adding or subtracting either 1 or  $\frac{1}{2}$  standard deviation from every variable in the quadrant corresponding to the chosen mode. The standard deviation is calculated from the entire data ( $\text{stdv} = 1$ ).

**Step 2: Generate intervened SAVAR targets.** We then determine the ground truth next step by running the prediction equation in SAVAR with the intervened data. Let  $\mathcal{X} \in \mathbb{R}^{\tau_{\max}} \times n \times d_x$  be the intervened data tensor replacing  $X$  in Eq. 3.2.1. We intervene on  $d_x/N$  variables, where  $N$  is the number of modes. For 4 modes, for example,  $d_x = 400$ , so we intervene on 100 variables.

**Step 3: Inference with intervened data.** The next step is to run inference on the trained models (MLP, CNN, LSTM, VAE) using the intervened data to generate the next step prediction.

**Step 4: Calculate RMSE on intervened targets and predictions.** For each model, we then calculate the RMSE between the targets calculated in Step 2, and the predictions generated in Step 3, over 1000 samples with random combinations of  $\text{stddev.}$ , time step, and mode.

### 3.3 Implementation Details

This section presents the technical details of the experimental setup, including model configurations, training procedures, and packages used. Hyperparameters are listed in Appx. A.

#### 3.3.1 Dataset generation

We generate datasets with different configurations using SAVAR. For number of modes, we select 4 and 16. We choose perfect square values in order to create square grids with an equal number of modes per row as per column. The spatial resolution of each mode quadrant is  $10 \times 10$  pixels, therefore datasets with different numbers of modes have the same resolution (information density) but different sizes. 4 modes were chosen because it is the simplest setting ( $2 \times 2$ ), used in both the initial SAVAR benchmark [74] and the PICABU paper [25], which we use as our causal representation learning method. 16 modes ( $4 \times 4$ ) provides a step up in complexity from 4 modes while remaining sufficiently manageable. We generate data for 10,000 time steps, with a 90/10 train/test split.

We set the causal graph to have a maximum time lag  $\tau_{\max}$  of 5. Tibau et al. [74] benchmark several causal discovery methods on SAVAR datasets with causal graphs that have a maximum lag between 1 and 3 time steps. Hickman et al. [25] evaluate PICABU on a maximum of 5 time steps. They also evaluate the performance of PICABU as an emulator by assessing how well PICABU captures the power spectral density of ENSO, a well-known climate index which occurs on the scale of months and years [72]. Often, datasets work with each discrete time block representing 1 month. So in this case,  $\tau_{\max} = 5$  would represent a causal “memory” of five months. A time lag of 5 was chosen to ensure that PICABU can learn relationships over as large a time lag as possible, while not compromising on training efficiency or resolution. To compare results, we similarly choose a time lag of 5.

Link weights are drawn from a Beta distribution  $\beta(4, 8)$  with mean  $1/3$  and standard deviation 0.13, then normalized. This is chosen to be comparable with the work by Hickman et al. [25]. These choices in distribution, mean, and standard deviation bound link values between 0 and 1, ensuring most are of modest strength ( $1/3$ ), and allowing for some stronger connections while avoiding links that are too close to 1, which would be unrealistic.

This work uses a total of 8 datasets ( $2 \text{ modes} \times 4 \text{ difficulties}$ ) all generated with the same seed (seed=1). The data is a time series of 2D images on a square grid, where the values have been normalized to a mean of 0 and standard deviation of 1. The values range is 30.

### 3.3.2 Training the Emulators

The training function is consistent across all models. Optimization is performed using the Adam optimizer with weight decay, which incorporates L2 regularization to mitigate overfitting. The learning rate follows a decay schedule throughout training. The loss function is the Mean Squared Error (MSE) between predictions and targets. Detailed hyperparameter settings are provided in the appendix Tab.A.1. The nonlinear activation function used in all models is the Rectified Linear Unit (ReLU, [44]). The MLP, CNN, and LSTM implementations follow the PyTorch examples [55].

**MLP** [59]. Composed of two fully-connected neural network layers, the first one followed by a nonlinear activation function and dropout for regularization.

**CNN** [36]. Consists of three convolutional layers followed by three fully connected layers, with max pooling and dropout for regularization.

**LSTM** [26]. Single LSTM layer paired with a two-layer MLP decoding layer. Using more than one LSTM layer led to significant overfitting.

**VAE** [32, 25]. PICABU is an encoder-decoder network based on a Variational Autoencoder (VAE), with a CNN encoder and decoder, and a normal-distributed latent. The latent representation is constrained to be a sparse causal graph. For more detail, see Section 2.3.3. In PICABU, the latent graph can be rolled out into the future and then decoded, which leads to a prediction of the next timestep for two-dimensional data. For our VAE experiment, we turn off the sparsity, orthogonality, and acyclicity constraints, thereby disabling the causal learning property of the model. It thus becomes a standard temporal VAE model, where the latent space is just a compressed representation of the data instead of having causal interpretation. We used a linear transition model, as opposed to a non-linear one.

### 3.3.3 Learning the latent causal graphs

#### Avg-PCMCI

The Gaussian weights used for dimensionality reduction are initialized using a Normal distribution. We implement PCMCI+ from the **Tigramite** package, a causal inference package for time series datasets. [62]. We create the dataframe that will be input into PCMCI+ with the analysis mode set to “multiple”, enabling the multidata setting. We initialize PCMCI+ with the ParCorr conditional independence test. We run PCMCI+ with  $\tau_{max} = 5$  and  $\tau_{min} = 1$  to disable instantaneous connections.

#### PICABU

Sparsity thresholds and hyperparameters for PICABU are reported in the appendix Tab.A.6

We test PICABU’s performance using both linear and nonlinear autoencoder architectures, combined with transition models that do and do not use parameter sharing. The linear autoencoder, which performs simple matrix multiplications, successfully converged on all but two datasets. On these two datasets (4 modes with medium easy and medium hard difficulty settings), training the linear model results in latent embeddings that explode numerically. We try to resolve the errors using strategies such as gradient clipping, but the issues persists. Due to time constraints, we use the nonlinear autoencoder variant for these two datasets. The nonlinear autoencoder uses neural network-based encoding and decoding with nonlinear activation functions.

Overall, we find that PICABU is very sensitive to hyperparameters. To obtain comparable results, we use hyperparameters similar to the ones used by Hickman et al. [25], as they also

test PICABU’s recovery of the ground truth causal graph on SAVAR.

### 3.3.4 Graph comparison metrics

We implement the SHD and causal metrics Parent-AID and OSet-AID from the `Gadjid` package for Causal Structure Learning [24]. The F1 score follows the equation described in the methods section 3.2.4.

### 3.3.5 Emulator performance metrics

All experiments are run with a seed value of 1. We use the `rfft` function from the Numpy package to compute the one-dimensional Discrete Fourier Transform (DFT) [22]. Interventional values for the Intervention RMSE are selected from  $[\pm 1, \pm \frac{1}{2}]$ . All other configurations for this experiment have been described above.

# Chapter 4

## Results

This section presents the main findings of our study. Experiment 1 validates the causal discovery methods and assesses correlations among the graph comparison metrics and model performance metrics respectively. In the following experiments, we assess correlation of the causal metrics and structural metrics in comparison to the model performance metrics: [RMSE](#) (Experiment 2), [LSD](#) (Experiment 3), Interventional RMSE (Experiment 4).

For brevity, we will use the term [Avg-PCMCI](#) to refer to the method that applies weighted-averaging dimensionality reduction followed by Multidata-[PCMCI+](#) causal discovery [60, 62] to learn the causal graph on 1000 samples of 20 autoregressive rollouts. One of the structural metrics included in this study is [SHD](#), however, as results were similar to [F1](#) score, and related works primarily focused on [F1](#) score (see 2.5), [SHD](#) plots have been moved to the appendix B.

### 4.1 Experiment 1: Validation of causal discovery and causal representation learning methods

#### 4.1.1 Recovery of ground truth

Our results demonstrate that [PICABU](#) substantially outperforms [Avg-PCMCI](#) in recovering the ground truth causal graph generated by [SAVAR](#) (Table 4.1 and Fig. 4.1. [PICABU](#) typically exhibits higher precision relative to recall (value between 0.95-1.0) when the exact structure is not fully recovered.

The [Avg-PCMCI](#) method may struggle for several reasons. (1) The dimensionality reduction step, which takes a weighted average with Gaussian weights of each mode quadrant may not accurately represent the signal of each mode, although it disentangles each mode accurately. (2) Performance of [PCMCI+](#) over long (1000) time steps yielded better [F1](#) score performance (ex. [F1](#) score of 1.0 on [SAVAR](#) data with 4 modes and medium easy difficulty). However, when generating autoregressive rollouts from model predictions, the residual compounds resulting in diminished values with successive steps, so we are limited in the number of time steps we could generate, which was selected to be 20. Future work could focus on improving the inference process, or tuning for the optimal number of rollouts.

To ensure comparability of the methods in Experiment 1 with subsequent experiments, we therefore have to structure the training data in a similar manner (i.e., 1000 samples of 20 time steps). As a result, the individual samples may have been too limited to represent the accurate propagation of causal information. We note that the reference causal graph was learned using the test data.

(3) We also observe that multi-[PCMCI+](#) identifies more links than there are true connections. We thus filter results with the highest learned weights, keeping only the expected number of links (obtained from the ground truth causal graph). This filtering method often discards true links and keeps spurious ones, due to the confidence values assigned by multi-[PCMCI+](#). Given these limitations, we exclude [PCMCI+](#) from further analysis, focusing instead on the more reliable [PICABU](#) method that better captures the underlying causal dynamics.

#### 4.1.2 Correlation analysis between metrics

Before comparing the graph metrics against model performance, we first need to examine correlations between the metrics themselves to determine whether the metrics are interchangeable.

| Modes | Difficulty | F1 Score    |           |
|-------|------------|-------------|-----------|
|       |            | PICABU      | Avg-PCMCI |
| 4     | Easy       | <b>1.0</b>  | 0.55      |
| 4     | Med-Easy   | <b>1.0</b>  | 0.14      |
| 4     | Med-hard   | <b>1.0</b>  | 0.56      |
| 4     | Hard       | <b>0.96</b> | 0.47      |
| 16    | Easy       | <b>1.0</b>  | 0.15      |
| 16    | Med-Easy   | <b>1.0</b>  | 0.04      |
| 16    | Med-hard   | <b>0.98</b> | 0.11      |
| 16    | Hard       | <b>0.89</b> | 0.11      |

Table 4.1: **Recovery of the ground truth causal graph from SAVAR using PICABU and Avg-PCMCI.** F1 score is the harmonic mean between precision and recall. PICABU recovers the ground truth graph perfectly on 5/8 datasets, while Avg-PCMCI fails to recover any perfectly. We observe that the difficulty setting (easy, medium easy, medium hard, hard) which dictates the number of links in the causal graph affects how well the graph can be learned by the different methods. Avg-PCMCI works poorly in comparison to PICABU, Interestingly, performance for Avg-PCMCI is lowest on the medium easy setting for both modes, whereas we would have expected that performance would be lowest on the hard setting.

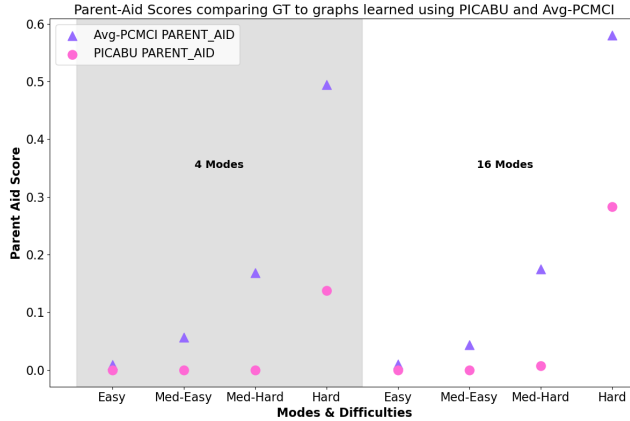
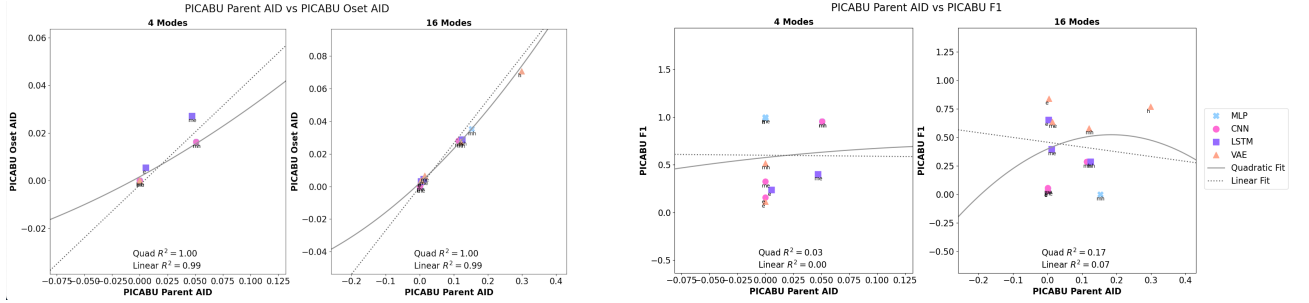


Figure 4.1: **Parent-AID scores for recovery of ground truth causal graph using PICABU compared to Avg-PCMCI.** Compared to Table 4.1 which looks at F1 score, we observe a more expected relationship between difficulty setting and the causal comparison metric Parent-AID, where values increase with increased difficulty.

Certain model runs are omitted from subsequent analysis for robustness: (1) PICABU runs into errors learning the causal graph from the VAE trained on four modes with difficulties medium easy and hard and time constraints prohibited further investigation; (2) The MLP on 16 modes results in extreme outliers (orders of magnitude difference) for model performance metrics. Values are displayed in the figures, but are omitted from correlation calculations. Fig. 4.6 provides further justification for this omission - we can see that the FFT coefficients for the MLP trained on 16 modes greatly diverge from the ground truth and other models. **We find no correlation between the causal metric Parent-AID and structural metrics** like F1 score in Fig. 4.2b or SHD in Fig. B.3. A strong correlation might otherwise have indicated redundancy and suggested similar correlations with the model performance metrics, which we do not observe. As a reminder, F1 score balances precision and recall. SHD counts incorrect edges and is not normalized, leading to higher values for graphs with more links (e.g., 16 modes vs. 4 modes).

**We do, however, find strong correlation between Parent-AID and OSet-AID,** which suggests that the parent sets are similar to the optimal adjustment sets, rendering the

use of the more sophisticated OSet-AID redundant. For the remainder of the work, we use Parent-AID as the causal metric we assess.



(a) **Parent-AID vs. OSet-AID.** The two causal metrics are highly correlated, so we use Parent-AID for the remainder of the analysis.

(b) **Parent-AID vs. F1.** No correlation is observed between Parent-AID and F1 score on PICABU for either 4 or 16 modes.

Figure 4.2: **Correlation between metrics.** Correlations between the causal metrics, and between the causal metrics and structural metrics. The metrics compare the PICABU-learned causal graphs between the emulators and SAVAR data. The labels **e**, **me**, **mh**, **h** refer to the difficulty setting of the SAVAR dataset that the model was trained on: easy, medium easy, medium hard, and hard respectively.

### 4.1.3 Correlation analysis between model performance metrics

To stay consistent with the previous section, we analyze the correlations among the model performance metrics.

We expected a positive correlation between these two metrics; models with low RMSE under normal conditions should also have lower RMSE under intervention. **Surprisingly, we observe a strong negative correlation between RMSE and intervention RMSE.** A possible interpretation for the strong negative trend seen in Fig. 4.3 could be that on datasets with “hard” difficulty, RMSE is higher (as expected). The hard setting increases causal connectivity in the latent graph governing the dynamics, so the output may be less affected by interventions. The models trained on the hard case will thus have lower intervention RMSE but higher RMSE.

As a reminder, LSD is a measure of how well the model captures spectral frequencies of the data. **We observe a medium positive correlation between RMSE and LSD on 4 modes, and a strong positive correlation on 16 modes.** This is in line with our expectations. A model that performs well on LSD likely has good RMSE (for both, lower values are better), however it is not guaranteed that a model with good RMSE will have a good LSD score due to the potential for overfitting. As expected from observing Fig. 4.3 and Fig. 4.4a, **there is a moderate negative relationship between intervention RMSE and LSD.**

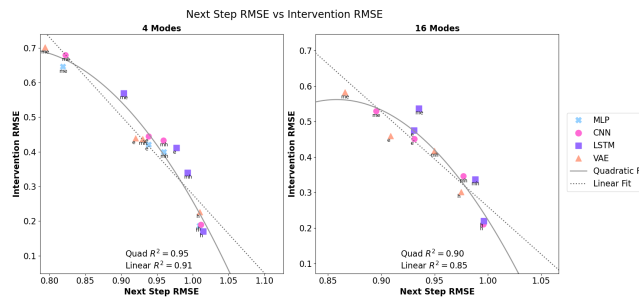


Figure 4.3: **Next step RMSE vs next step RMSE with intervention.** Next step RMSE values are obtained in Experiment 2, and next step RMSE values with intervention are obtained in Experiment 4. Surprisingly, we observe a strong negative correlation between RMSE and intervention RMSE.

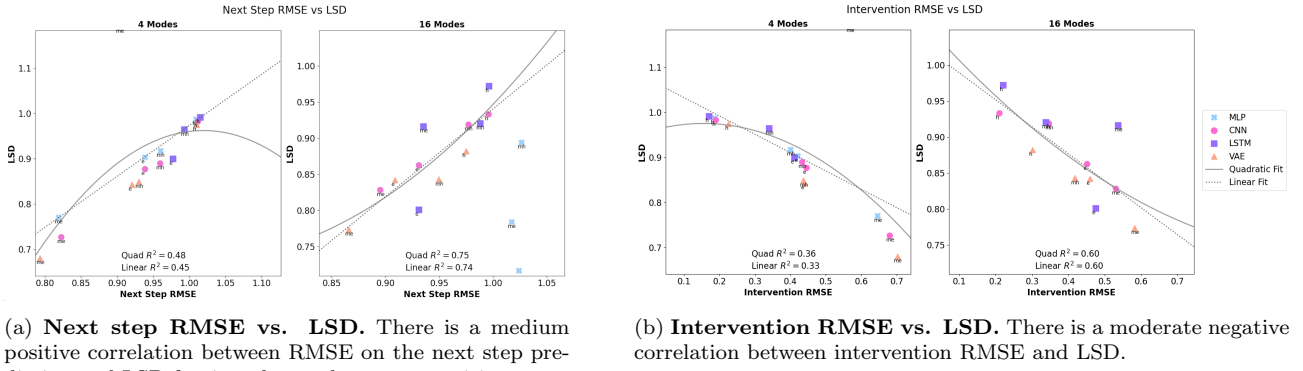


Figure 4.4: **Performance Metric Correlation:** A correlation between the different model performance metrics suggests that they may be capturing overlapping aspects of model behaviour.

## 4.2 Experiment 2: RMSE

| Mode        | Average range | Average stdev |
|-------------|---------------|---------------|
| Easy        | 14.090        | 0.356         |
| Medium easy | 15.388        | 0.457         |
| Medium hard | 8.223         | 0.273         |
| Hard        | 4.583         | 0.152         |

Table 4.2: Summary of range and standard deviation by mode, averaged across all model one-step outputs.

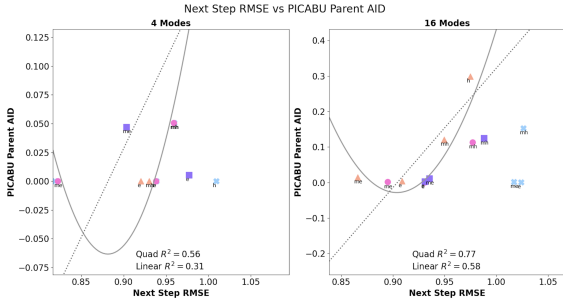
Experiment 2 compares RMSE on next-step prediction with the graph evaluation metrics Parent-AID and F1 score, applied to graphs learned using PICABU. See Appendix B for Avg-PCMCi results. Table 4.2 summarizes the model statistics on next-step outputs. The training and target SAVAR data has an average range of 30 with a standard deviation of 1.0. However, the average range and standard deviation across all model outputs is reduced by approximately 1/3, to 10.57 and 0.31 respectively. Across all models, the average output bias is approximately -1.0, and the mean is 0. This scaling explains why the variation of outputs observed in Fig. 3.3 appears muted. It is unclear why this occurs, but it likely explains the signal decay observed in long autoregressive rollouts.

**We observe no correlation between RMSE and F1 score** (Fig. 4.5b or SHD B.7, which aligns with our expectations. **There is a moderate linear correlation between the causal metric Parent-AID and RMSE.** Although this correlation is not strong, it also aligns with our expectations, as we might expect low Parent-AID values to correspond with low RMSE, but low RMSE values alone do not guarantee low Parent-AID scores, likely due to overfitting. This reinforces the view that RMSE alone is an imperfect predictor of the underlying model dynamics.

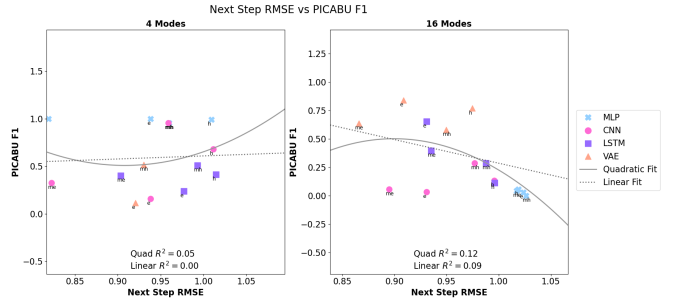
## 4.3 Experiment 3: Power spectral density analysis

We compare LSD to the graph evaluation metrics Parent-AID and F1 score, applied to graphs learned using PICABU. Results for Avg-PCMCi are provided in Appendix B.

**There is no observable relationship between LSD and structural metrics such as SHD and F1 score, but a slightly stronger  $R^2$  correlation emerges with the causal metric Parent-AID for 16 modes.** We argue that LSD offers deeper insight into the underlying system dynamics compared to RMSE, because it reveals patterns in the underlying



(a) **Next step RMSE vs. Parent-AID.** We observe a moderate positive correlation.



(b) **Next step RMSE vs. F1 Score.** No correlation is observed

Figure 4.5: Comparing model performance on RMSE next step prediction to the causal metric Parent-AID and structural metric F1 score applied to the causal graphs learned by PICABU on each emulator.

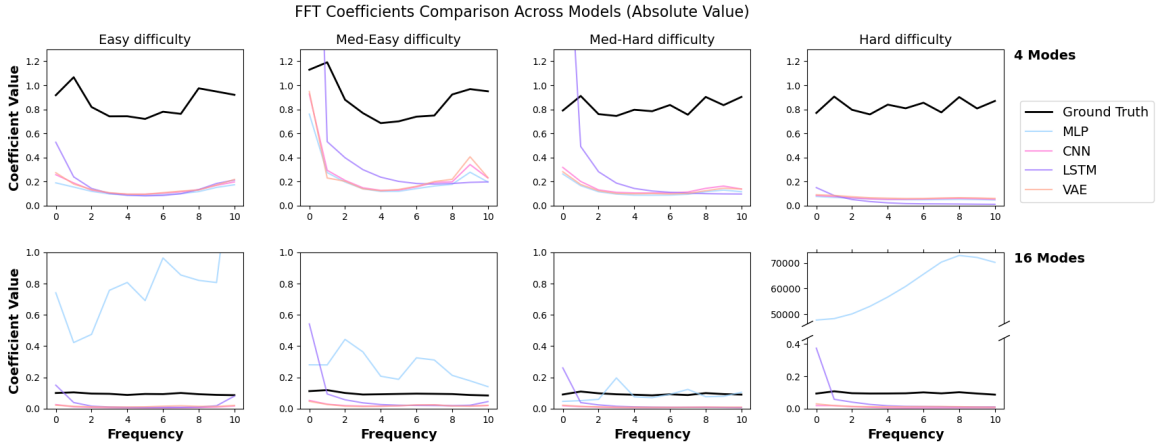


Figure 4.6: **FFT Coefficient comparison across models.** Coefficient values obtained from the Fast Fourier Transform (FFT) applied to 20-step autoregressive rollouts of each model. These are the values prior to applying LSD. Although the magnitude of the coefficient values for most models is lower than the ground truth, the power spectra of many models maintain a consistent shape with respect to the target. On 16 modes, the MLP diverges significantly from the reference, reflecting poor fidelity in representing the underlying dynamics.

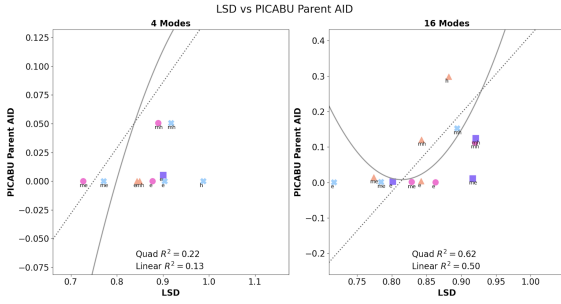
frequency spectrum of the data. Thus, a low-to-moderate correlation between LSD and Parent-AID is encouraging, suggesting that LSD reflects aspects of the causal structure better than structural metrics.

The correlation between LSD and Parent-AID on 4 modes is very low. Some models learn causal graphs that have perfect agreement with the reference data in terms of causal effect identification (Parent-AID). This does not necessarily mean that the graphs are structurally identical, just that their identification formulas are in agreement. Future work might benefit from more complex datasets with more modes, to better assess trends in settings where the Parent-AID score is not perfectly zero.

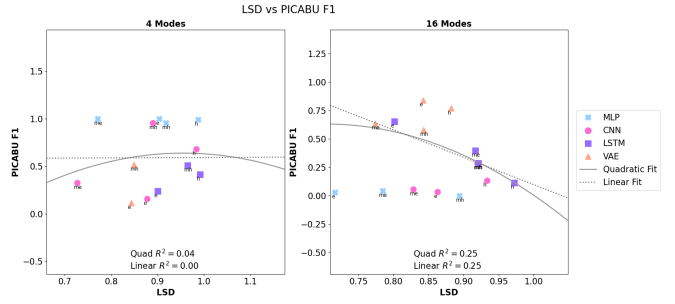
## 4.4 Experiment 4: Interventions

To evaluate the robustness of the models, we apply interventions to the input data of each trained model by modifying selected mode values at specific input time steps. We then use RMSE to compare the model’s next-step predictions on the intervened data to the known [SAVAR](#) ground truth generated from the intervened data. Since the model is already trained, the causal graph does not change. We expect that a model that has accurately learned the causal graph will exhibit robustness to interventions, reflected by lower intervention RMSE.

Consistent with previous results, structural metrics show no correlation with intervention RMSE (see Fig. 4.8b). Interestingly, we observe a negative correlation



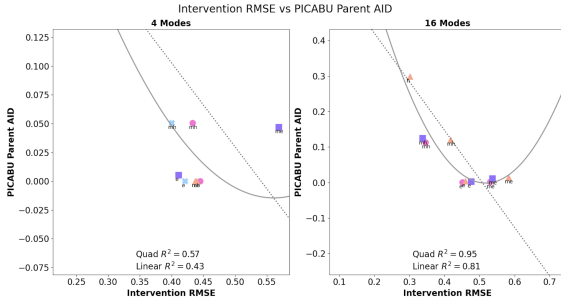
(a) **LSD vs Parent-AID.** Low-to-moderate positive correlation is observed.



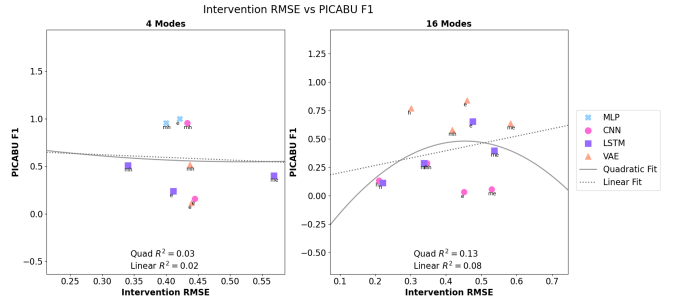
(b) **LSD vs. F1 Score.** While no correlation is observed in 4 modes, a low negative correlation is observed in 16 modes

Figure 4.7: Comparison of model fidelity in capturing system dynamics: least squares distance on FFT coefficients from 20-step autoregressive rollouts versus the fidelity of causal graphs learned using PICABU, measured by the Parent-AID metric. An F1 score of 1 indicates perfect structural alignment with the reference graph) whereas a lower LSD score is better.

with Parent-AID, contrary to the positive correlation initially expected. This negative correlation, however, aligns with the negative relationship between standard RMSE and intervention RMSE observed in Fig. 4.3.



(a) **Comparison of intervention RMSE to PICABU Parent-AID metric for models trained on 4 modes (left) and 16 modes (right).** For both modes, there is a negative quadratic and linear trend between intervention RMSE and Parent-AID.



(b) **Comparison of intervention RMSE to PICABU F1 score for models trained on 4 modes (left) and 16 modes (right).** No significant correlation is observed between intervention RMSE and F1 score across either mode setting suggesting structural accuracy of the learned graphs does not predict robustness to intervention.

Figure 4.8: Comparison of model robustness to interventions, as measured by intervention RMSE, against fidelity of the underlying causal graph (learned using PICABU) for different model architectures and mode settings.

# Chapter 5

## Conclusion

In this work, we set out to find metrics that tell us whether climate emulators have implicitly learned spurious correlations, or true dynamics underlying the data. We provided an evaluation and test framework for CME in a synthetic setting where we have access to the ground truth causal factors, allowing for rigorous quantitative analysis. We showed that there is greater correlation between performance and causal metrics than performance and structural metrics. In other words, we found that causal representation learning in conjunction with causal metrics (specifically PICABU + Parent-AID) is a better indicator of model performance. We found comparable trends on F1 score and Parent-AID for Avg-PCMCI (Appx. B).

**Limitations.** (1) The framework was set up to run different seeds, but due to time constraints we ran experiments with only one seed. Running experiments on more seeds and more modes would allow us to: (a) run significance statistical tests since conclusions cannot be based on correlation alone, and (b) disentangle the effects of the data difficulty and model type on the metrics. (2) We did not optimize hyperparameters of the causal discovery methods and used the parameters reported in prior literature. As mentioned in the main implementation section, PICABU is especially sensitive to hyperparameters and could have possibly benefited from hyperparameters better tuned to this scenario (possibly eliminating problems with NaNs and crashed runs). However, in the context of evaluating climate models, there is no ground-truth and it is not possible to tune PICABU without extensive manual evaluation. Furthermore, beyond basic regularization to prevent overfitting, we did not optimize the hyperparameters of the neural network emulators. As a result, the performance of the models is suboptimal, and we were limited in the complexity of the datasets we could use. (3) The methods we used to learn the causal graphs assume that there are no contemporaneous links. This is a simplifying assumption that may not be reflective of real-world climate dynamics [30]. (4) Compared to Avg-PCMCI+, PICABU does not learn link weights, which is important to understand the impact of different variables in the causal graph. We removed the weights learned by Avg-PCMCI+ to make our results comparable to PICABU.

**Future Work.** (1) Due to the computational limits of numerical climate models, and the limitations of performing real-world experiments on the climate, interventions are not always possible. Having a proxy (obtained through CME) for model robustness to interventions would be highly valuable, thus further study is warranted. The difference in range between RMSE and intervention RMSE suggests that the intervention step causes homogeneity in the next time step. In hard mode, where variables are highly interconnected, homogenization may lower RMSE as model predictions align with this smoothed outcome. In easy mode, with fewer interconnections, interventions disrupt the system more, resulting in a higher RMSE. Future work could focus on understanding how interventions in the data affect the next time step step, to design interventions that are proportional to the connectivity of the underlying causal graph. (2) Methods and metrics should be explored that account for the weights of the causal links. (3) We were able to establish interesting correlations in a “lab setting” with SAVAR, but a crucial next step would be to extend our methods to “real-world” climate-science applications, using ESMs such as NorESM [67] and emulators such as ACE2 [80]. While further investigation is required to assess the robustness of these findings, this work represents a critical first step toward identifying metrics that characterize a climate emulator’s “internal understanding”, offering a promising framework for the climate community to adapt and extend.

# Acronyms

- CDSD** Causal Discovery with Single-Parent Decoding, see section 2.3.3. 17
- CME** Causal Model Evaluation, see section 2.5. 40
- CMIP** Coupled Model Intercomparison Projects, see section 2.1.1. 21
- CNN** Convolutional Neural Network. i, 31
- CPDAG** Describes a Markov equivalence class of DAGs, has both directed and undirected edges. 11, 14, 20
- CRL** Causal Representation Learning, see section 2.3.2. 16
- DACE** Distance Average Causal Effect, see section 2.1.1. 29
- DAG** directed acyclic graph, see section 2.3.1. 26
- DL** Deep Learning. 1, 3
- ENSO** El Niño-Southern Oscillation, see section 2.1.2. 5, 31
- ESM** Earth System Model, see section 2.1.1. 1, 40
- F1** F1 Score, see section 2.4.1. i, 23, 34, 40
- FFT** Fast Fourier Transform. 23, 35
- LSD** Least Squares Distance, see section 2.1.5. i, 23, 34
- LSTM** Long Short-Term Memory Network. i, 31
- MEC** Markov equivalence class, see section 2.3.1. 11, 15, 20
- MLP** Multi-Layer Perceptron. i, 31, 35
- Parent-AID** Parent-set based Adjustment Identification Distance, see section 2.4.2. i, 23, 35, 40
- PCMCI** Peter and Clark Momentary Conditional Independence Plus, see section 2.3.2. i, 1, 21, 23, 34
- PICABU** Physics-Informed Causal Approach with Bayesian Uncertainty, see section 2.3.3. i, 1, 23, 34, 40
- RMSE** Root Mean Square Error, see section 2.1.5. i, 23, 34, 40
- SAVAR** spatially aggregated vector-autoregressive (model) - “a spatiotemporal stochastic climate model for benchmarking causal discovery methods for [teleconnections](#)”. See section 3.2.1. i, 1, 23, 34, 38
- SCM** Structural Causal Model, see section 2.3.1. 25

**SHD** Structural Hamming Distance, see section [2.4](#). [i](#), [23](#), [34](#)

**VAE** Variational Autoencoder. [i](#), [31](#), [35](#)

# Glossary

**Avg-PCMCI** The combination of dimensionality reduction (weighted averaging of mode regions using Gaussian weights) on samples of 20 time steps, followed by Multidata-PCMCI+ causal discovery. [23](#), [34](#), [40](#)

**climatology** Average conditions for a location at a particular time of year, often monthly mean values. [6](#)

**contemporaneous link** A link between variables where cause and effect are observed instantaneously at the same time step (lag 0). [16](#), [20](#), [29](#), [40](#)

**latent variable** Variables that are inferred from the data, rather than directly observed. In the context of this work, latents refer to variables that govern the underlying mechanisms in a system. [4](#), [25](#)

**teleconnections** Links between weather phenomena occurring at different locations on earth . [4](#), [41](#)

**v-structure** Triples in a DAG where two parent nodes point to the same child node, a collider, and the parents are not adjacent. [11](#)

# Appendix A

## Hyperparameters for experiment configuration

The following section lists non-standard hyperparameters. If a given hyperparameter for a model or method is not listed here, the default value for the given implementation are used. MLP, LSTM, and CNN model training parameters are shown in Tab.A.1.

| Hyperparameter             | MLP                | LSTM               | CNN                |
|----------------------------|--------------------|--------------------|--------------------|
| <i>Model Parameters</i>    |                    |                    |                    |
| Number of Layers           | 2                  | 1                  | -                  |
| Channels                   | -                  | -                  | [32, 64, 128]      |
| Kernel Sizes               | -                  | -                  | [3, 3, 5]          |
| FC Layers                  | -                  | -                  | [512, 256]         |
| <i>Training Parameters</i> |                    |                    |                    |
| Epochs                     | 100                | 100                | 100                |
| Batch Size                 | 128                | 128                | 128                |
| Learning Rate              | $1 \times 10^{-2}$ | $1 \times 10^{-3}$ | $1 \times 10^{-3}$ |
| Gamma ( $\gamma$ )         | 0.7                | 0.7                | 0.7                |
| Validation Split           | 0.2                | 0.2                | 0.2                |

Table A.1: MLP, LSTM, and CNN Hyperparameters

The hyperparameters for the PICABU VAE are shown in Tab.A.2.

| Hyperparameter                | Value     |
|-------------------------------|-----------|
| Ortho $\mu$ initialization    | $10^{-8}$ |
| Ortho $\mu$ factor            | 1         |
| Ortho $h$ threshold           | $10^8$    |
| Sparsity $\mu$ initialization | $10^{-8}$ |
| Sparsity $\mu$ factor         | 1         |
| Sparsity $h$ threshold        | $10^8$    |
| Acyclic $\mu$ initialization  | $10^{-8}$ |
| Acyclic $\mu$ factor          | 1         |
| Acyclic $h$ threshold         | $10^8$    |

Table A.2: VAE Hyperparameters

The size- and difficulty-independent hyperparameters for SAVAR are shown in Tab.A.3.

The PICABU standard and special case hyperparameters (only used for 4 modes - medium easy & medium hard) are shown in Tab.A.4 and Tab.A.5 respectively. Below that are the sparsity threshold parameters for PICABU for different modes in SAVAR in Tab.A.6.

| Hyperparameter                             | Value<br>4 Modes | Value<br>16 Modes |
|--------------------------------------------|------------------|-------------------|
| Number of modes, <code>d_z</code>          | 4                | 16                |
| Total data dimension, <code>d_x</code>     | 400              | 1600              |
| Horizontal dimension, <code>lon</code>     | 20               | 40                |
| Vertical dimension, <code>lat</code>       | 20               | 40                |
| Spatial resolution, <code>comp_size</code> | 10               | 10                |
| Items per column, <code>n_per_col</code>   | 2                | 4                 |

Table A.3: Savar Global Hyperparameters

| Hyperparameter              | Value  |
|-----------------------------|--------|
| <i>Training Parameters</i>  |        |
| Batch size                  | 128    |
| Eval batch size             | 128    |
| Learning rate               | 0.0003 |
| <i>Model Parameters</i>     |        |
| Number of layer mixings     | 0      |
| Position embedding dim      | 100    |
| Use gradient norm           | True   |
| <i>Optimizer Parameters</i> |        |
| Binarize transitions        | False  |
| <i>Rollout Parameters</i>   |        |
| Batch size                  | 995    |
| Number of particles         | 1      |
| Sub-particles               | 1      |
| Number of timesteps         | 20     |

Table A.4: PICABU Hyperparameters

| Hyperparameter           | Value |
|--------------------------|-------|
| <i>Model Parameters</i>  |       |
| Nonlinear Mixing         | True  |
| Number of hidden mixings | 8     |
| Number of layer mixings  | 2     |
| Number of hidden units   | 8     |
| Number of layers         | 2     |

Table A.5: PICABU Hyperparameters for non-linear transition models (used for 4 modes, difficulty `med_easy` and `med_hard`)

| Difficulty | Modes | Sparsity Threshold | General Equation                  |
|------------|-------|--------------------|-----------------------------------|
| Easy       | 4     | 0.05               | $\frac{N}{N^2\tau}$               |
|            | 16    | 0.0125             |                                   |
| Med Easy   | 4     | 0.1                | $\frac{2N}{N^2\tau}$              |
|            | 16    | 0.025              |                                   |
| Med Hard   | 4     | 0.15               | $\frac{3N}{N^2\tau}$              |
|            | 16    | 0.0375             |                                   |
| Hard       | 4     | 0.2                | $(N + N(N - 1))/(N^2\tau)$        |
|            | 16    | 0.10625            | $(N + \frac{N(N-1)}{2})(N^2\tau)$ |

Table A.6: PICABU Sparsity Thresholds by Difficulty and Number of Modes

# Appendix B

## Additional results

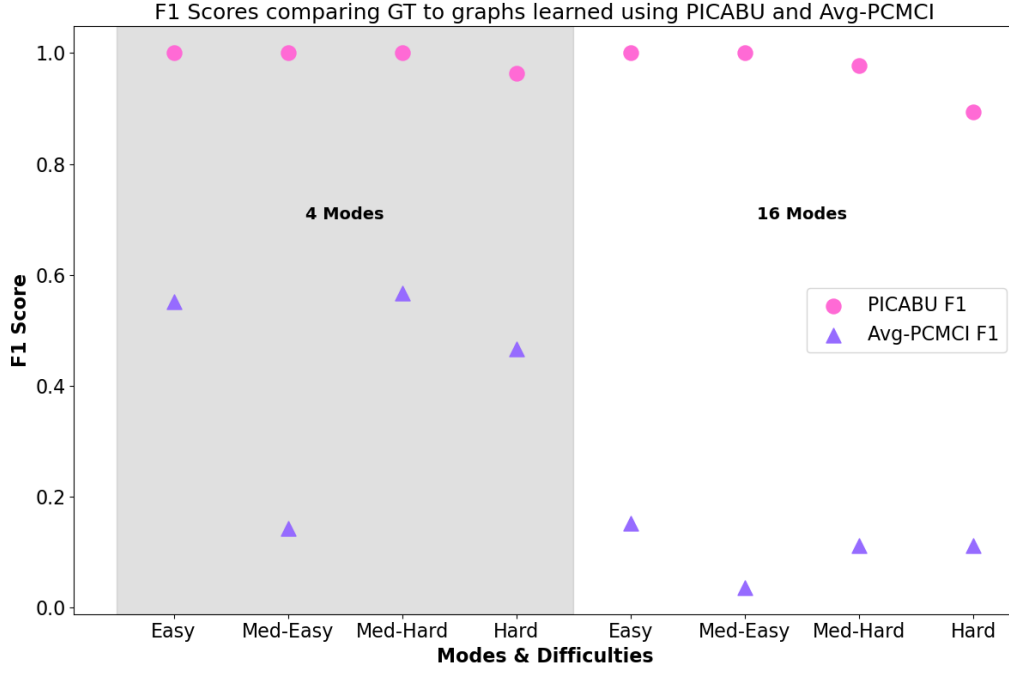


Figure B.1: **Experiment 1: F1 scores for recovery of ground truth causal graph using PICABU vs Avg-PCMCI.** Score of 1.0 means the graph structure was learned exactly. PICABU was able to recover the ground truth with high fidelity, whereas Avg-PCMCI could not.

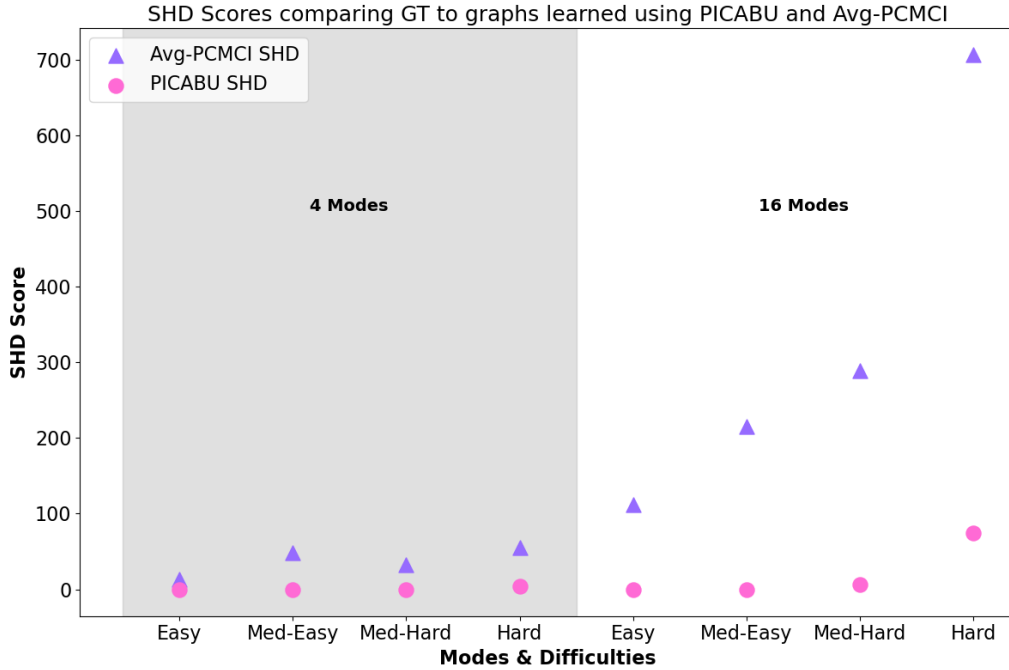


Figure B.2: **Experiment 1: SHD scores for recovery of ground truth causal graph using CRL (PICABU) vs CD (weighted mode averaging with multi-PCMCI+).** SHD counts the number of incorrect links between the learned graph and the ground truth. SHD is not normalized, so the metric scales with the number of links in the graph. As there are more expected links in a graph with 16 modes than 4 modes, and difficulty increases expected number of links. Although PICABU performed better than Avg-PCMCI, performance decreased with complexity of the dataset.

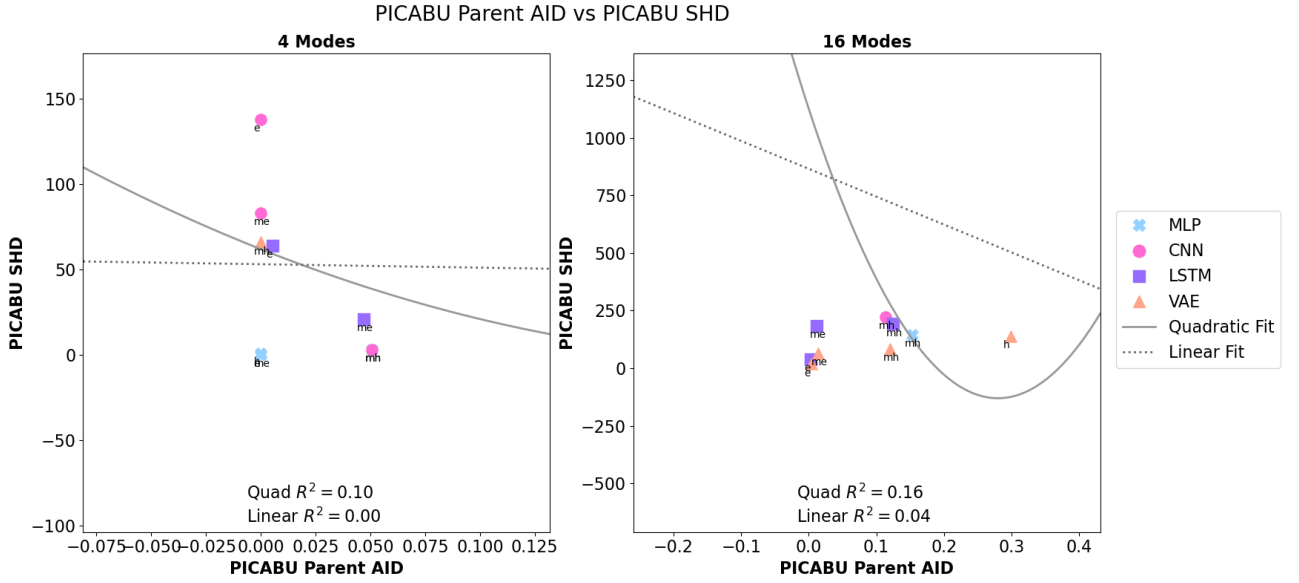


Figure B.3: **Experiment 1: Parent-AID vs SHD for PICABU.** There is no correlation between Parent-AID and SHD score on PICABU for either 4 or 16 modes

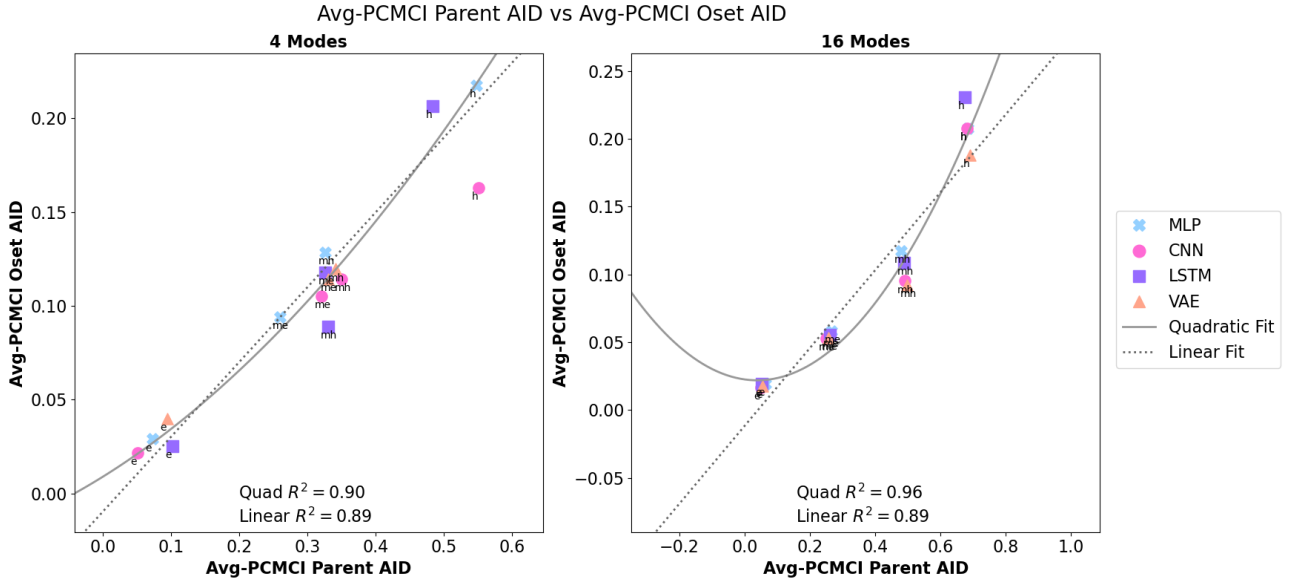


Figure B.4: **Experiment 1: Parent-AID vs OSet-AID for Avg-PCMCI.** There is a strong positive correlation between the causal metrics Parent-AID and OSet-AID applied to the causal graphs learned by Avg-PCMCI.

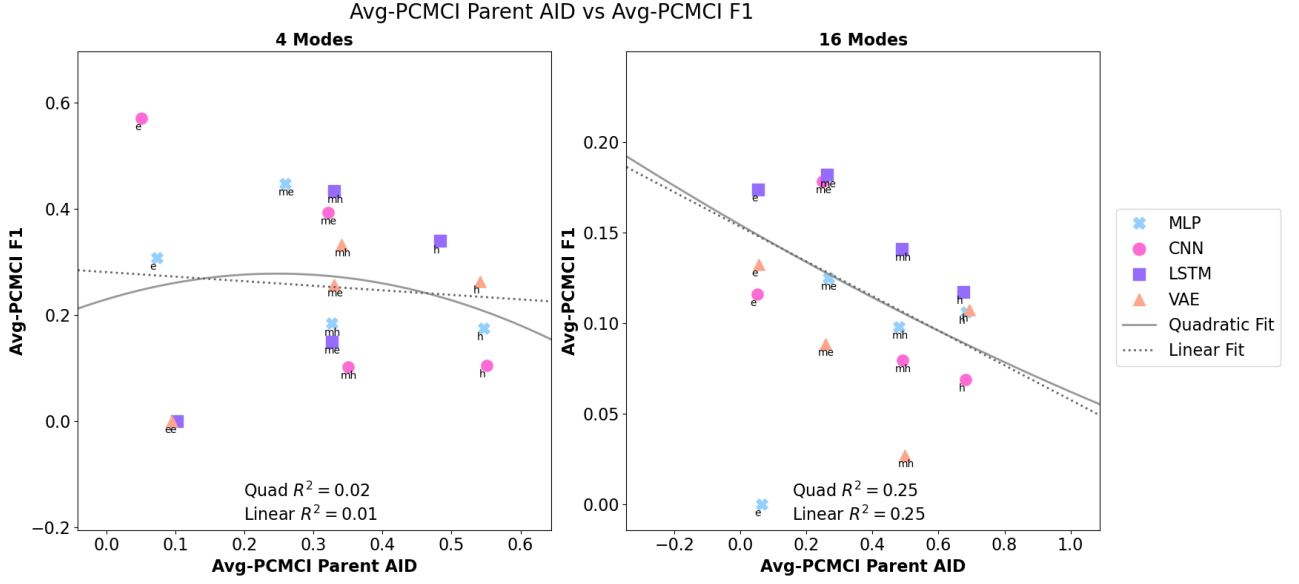


Figure B.5: **Experiment 1: Parent-AID vs F1 score for Avg-PCMCI.** There is a no correlation for 4 modes, and a weak negative correlation for 16 modes, between the causal metrics Parent-AID and OSet-AID applied to the causal graphs learned by Avg-PCMCI.

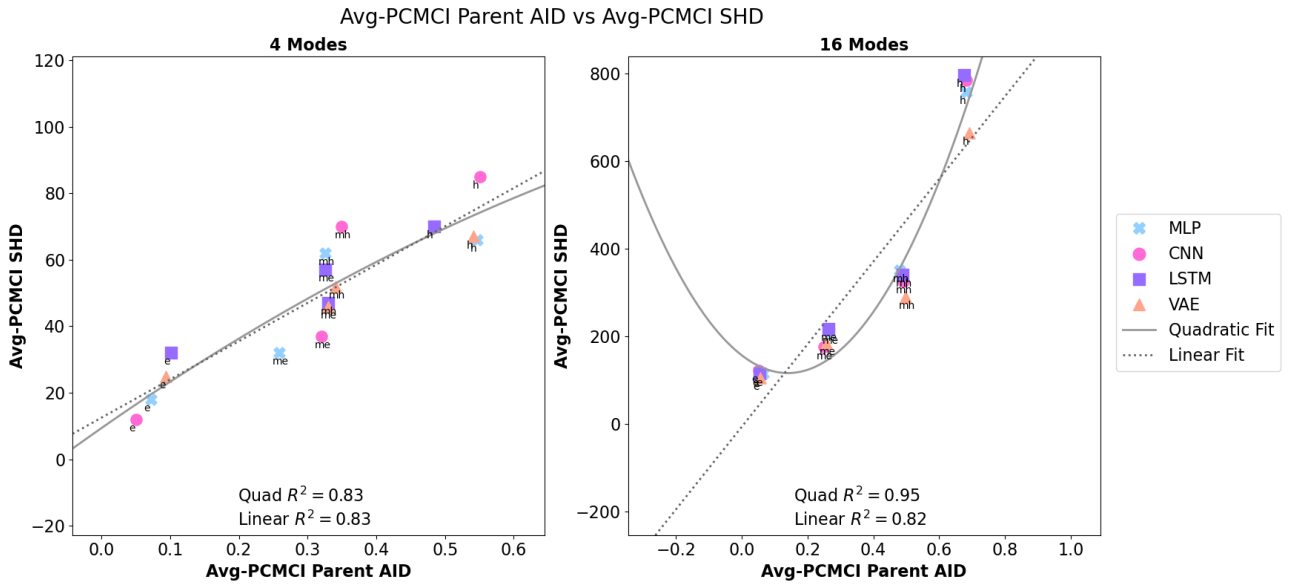


Figure B.6: **Experiment 1: Parent-AID vs SHD for Avg-PCMCI.** There is a strong positive correlation between the causal metrics Parent-AID and SHD applied to the causal graphs learned by Avg-PCMCI.

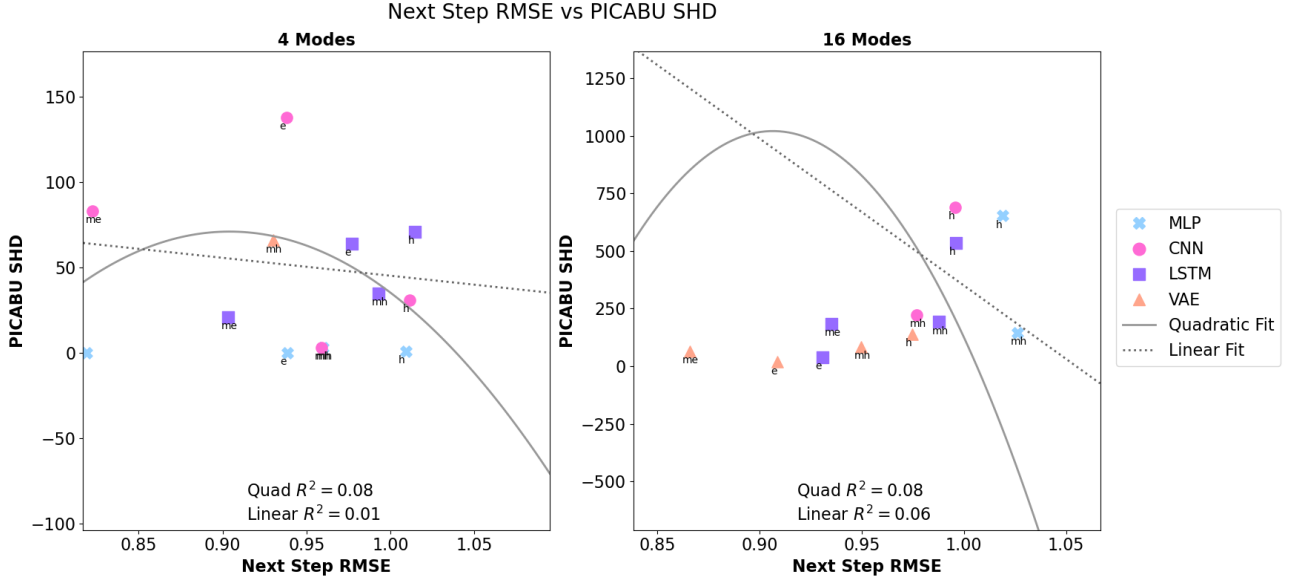


Figure B.7: **Experiment 2: Next step RMSE vs SHD for PICABU.** There is no correlation between the structural metric SHD applied to the causal graphs learned by PICABU and the emulator performance metric, next step RMSE.

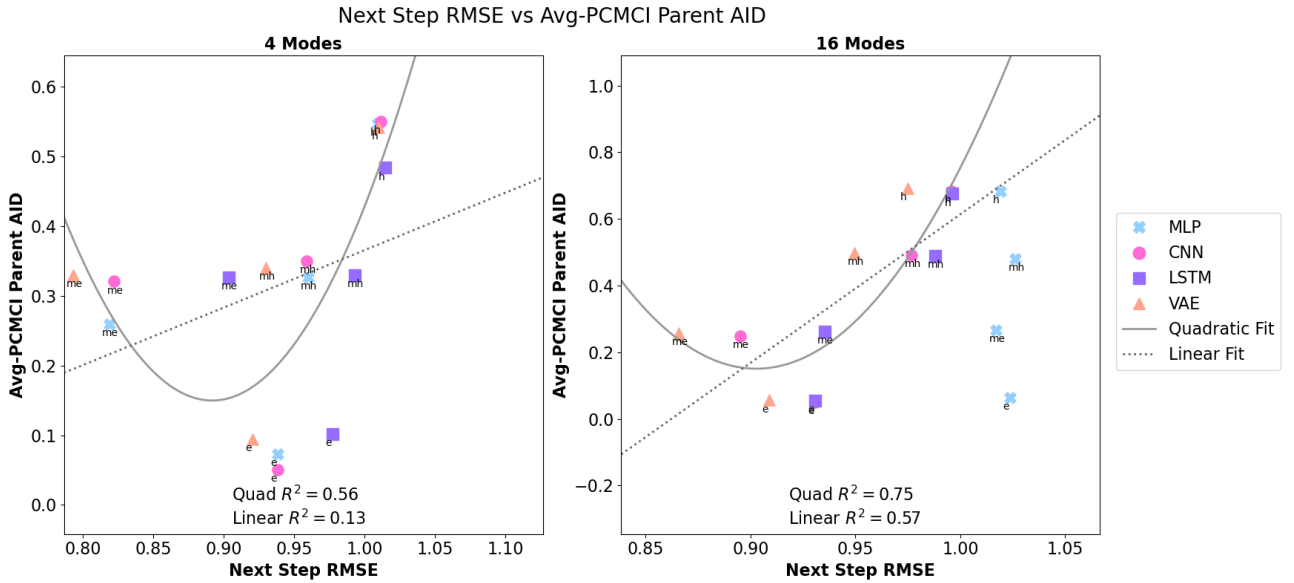


Figure B.8: **Experiment 2: Next step RMSE vs Parent-AID for Avg-PCMCI.** There is a moderate quadratic correlation, and a moderate-to-strong correlation on 16 modes between the causal metric Parent-AID applied to the causal graphs learned by Avg-PCMCI and the emulator performance metric, next step RMSE.

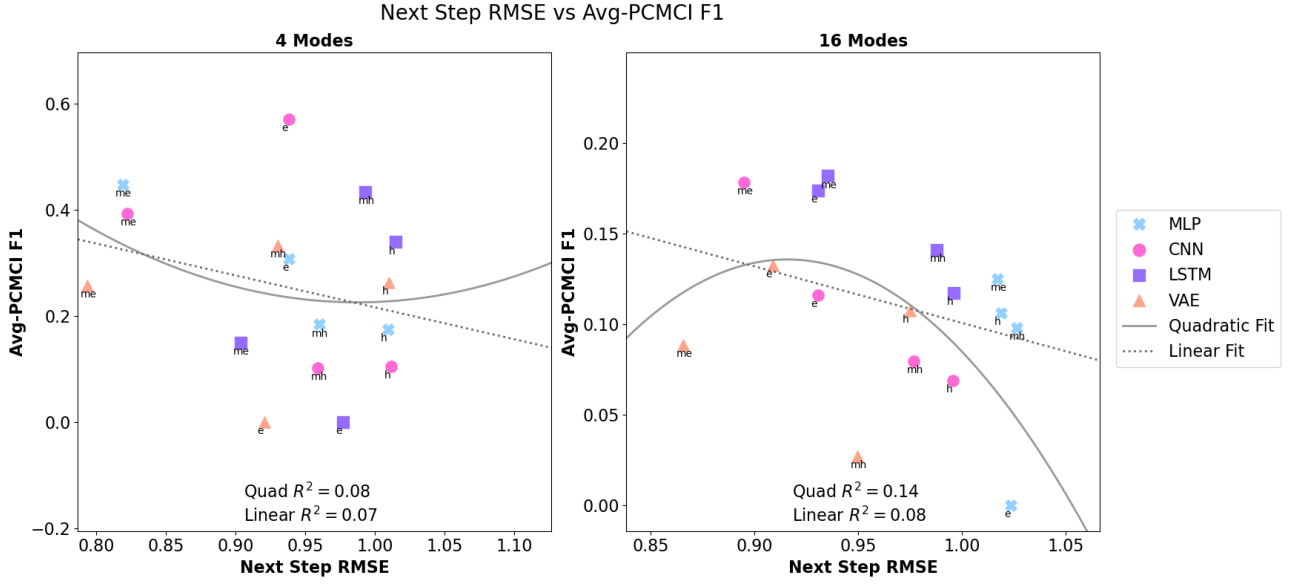


Figure B.9: **Experiment 2: Next step RMSE vs F1 for Avg-PCMCI.** There is no correlation between the structural metric F1 score applied to the causal graphs learned by Avg-PCMCI and the emulator performance metric, next step RMSE.

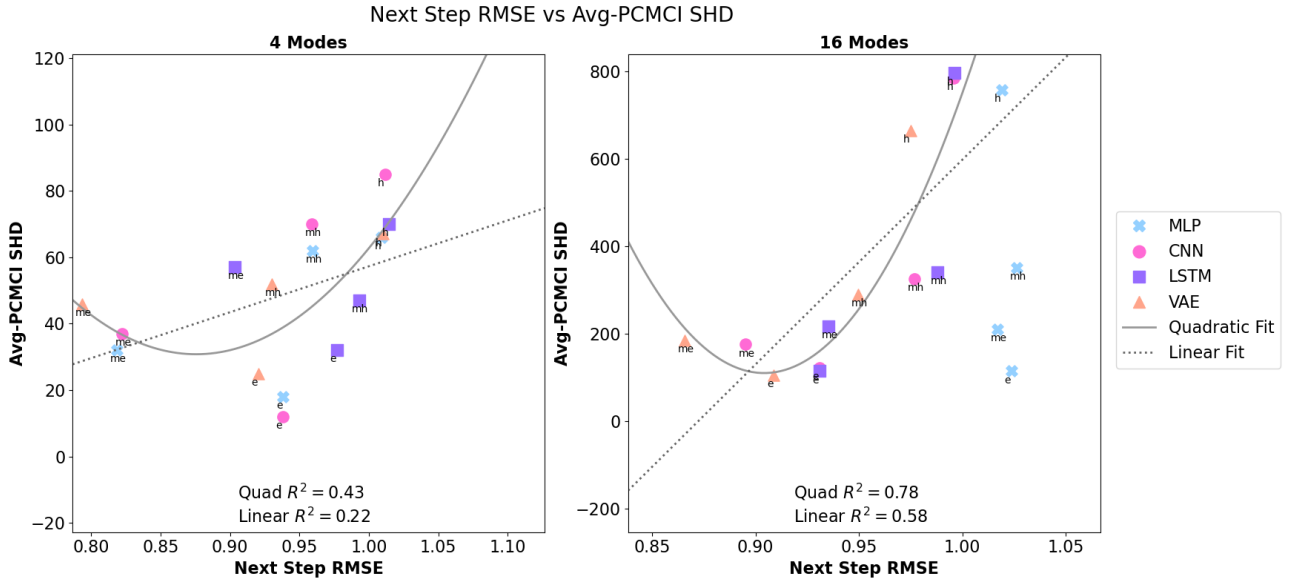


Figure B.10: **Experiment 2: Next step RMSE vs SHD for Avg-PCMCI.** There is a low positive correlation for 4 modes, and a moderate-to-high positive correlation for 16 modes between the structural metric SHD score applied to the causal graphs learned by Avg-PCMCI and the emulator performance metric, next step RMSE.

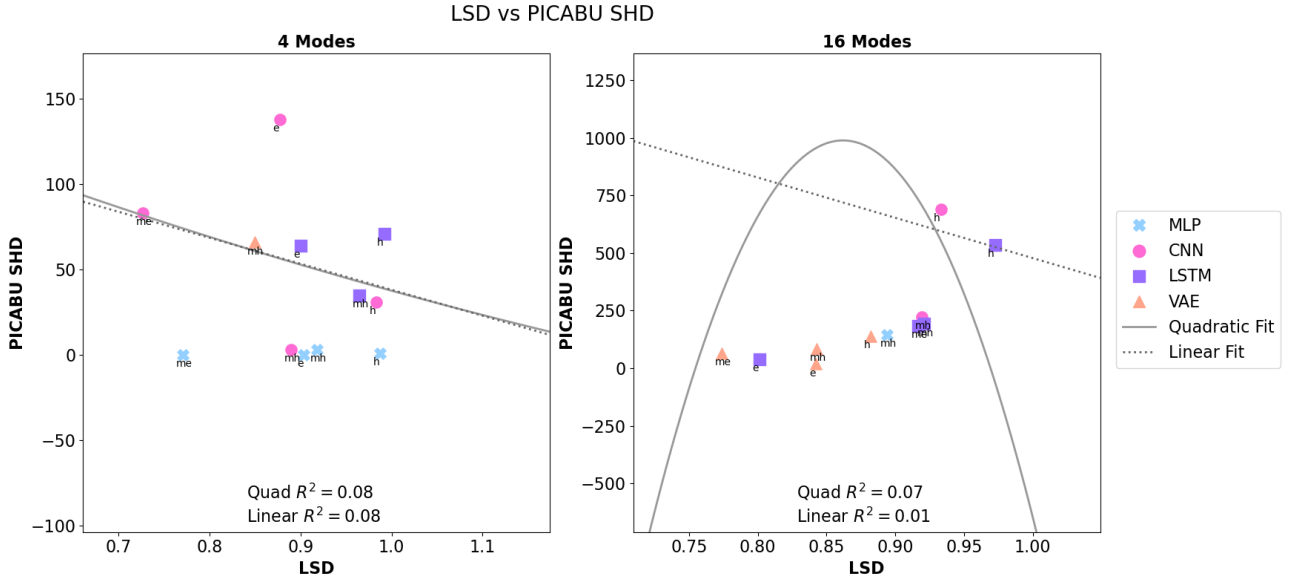


Figure B.11: **Experiment 3: LSD vs SHD for PICABU.** There no between the structural metric SHD applied to the causal graphs learned by PICABU and the emulator performance metric, LSD.

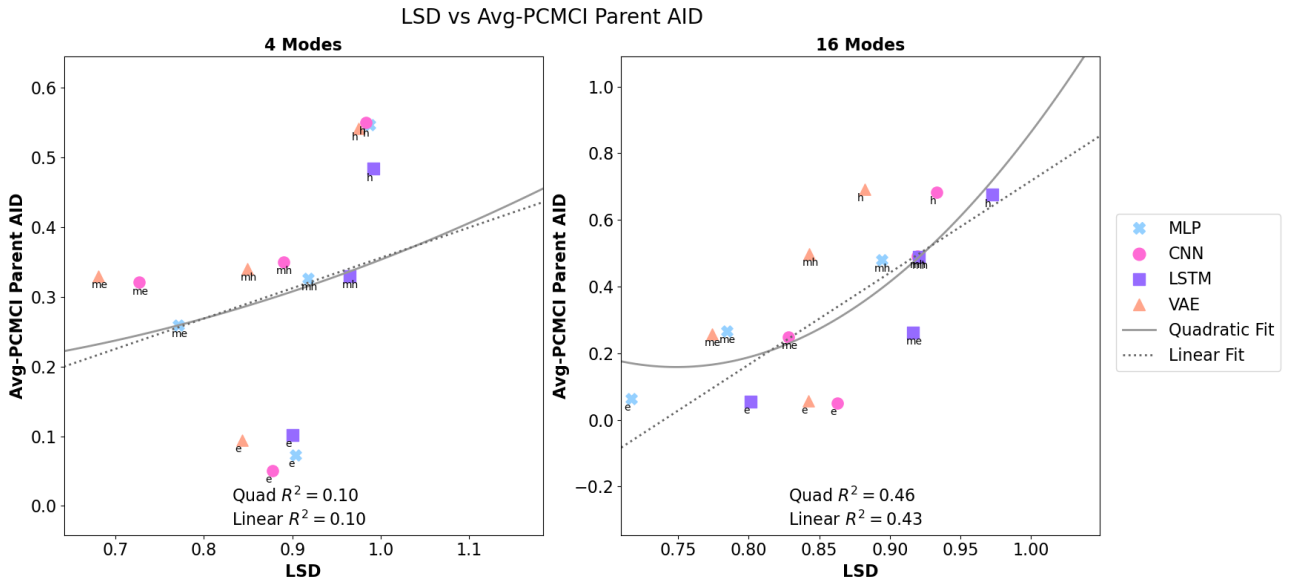


Figure B.12: **Experiment 3: LSD vs Parent-AID for Avg-PCMCI.** There no correlation for 4 modes, and a weak-to-moderate correlation for 16 modes between the causal metric Parent-AID applied to the causal graphs learned by Avg-PCMCI and the emulator performance metric, LSD.

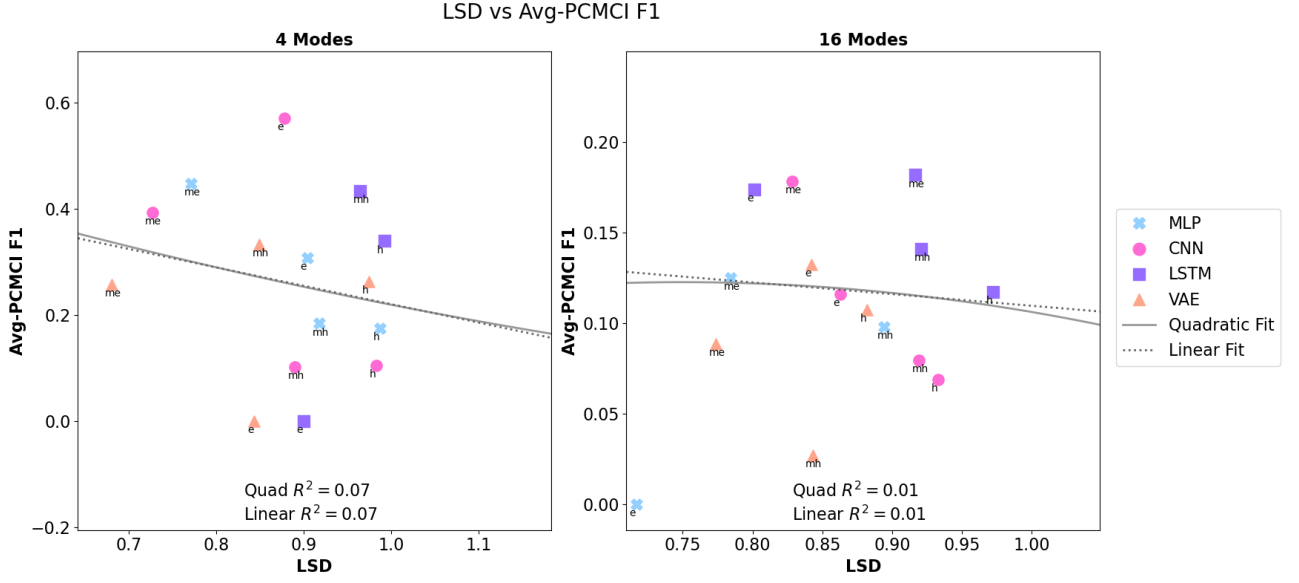


Figure B.13: **Experiment 3: LSD vs F1 Score for Avg-PCMCI.** There no correlation between the structural metric F1 score applied to the causal graphs learned by Avg-PCMCI and the emulator performance metric, LSD.

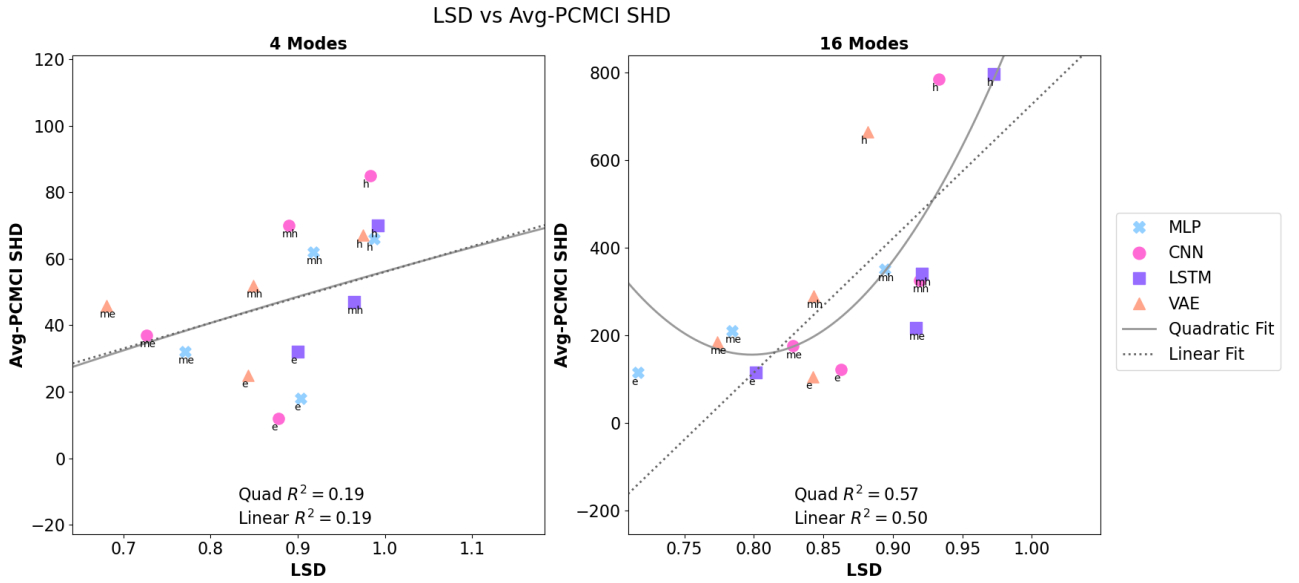


Figure B.14: **Experiment 3: LSD vs SHD for Avg-PCMCI.** There is a low positive correlation for 4 modes, and a moderate positive correlation for 16 modes between the structural metric SHD score applied to the causal graphs learned by Avg-PCMCI and the emulator performance metric, LSD.

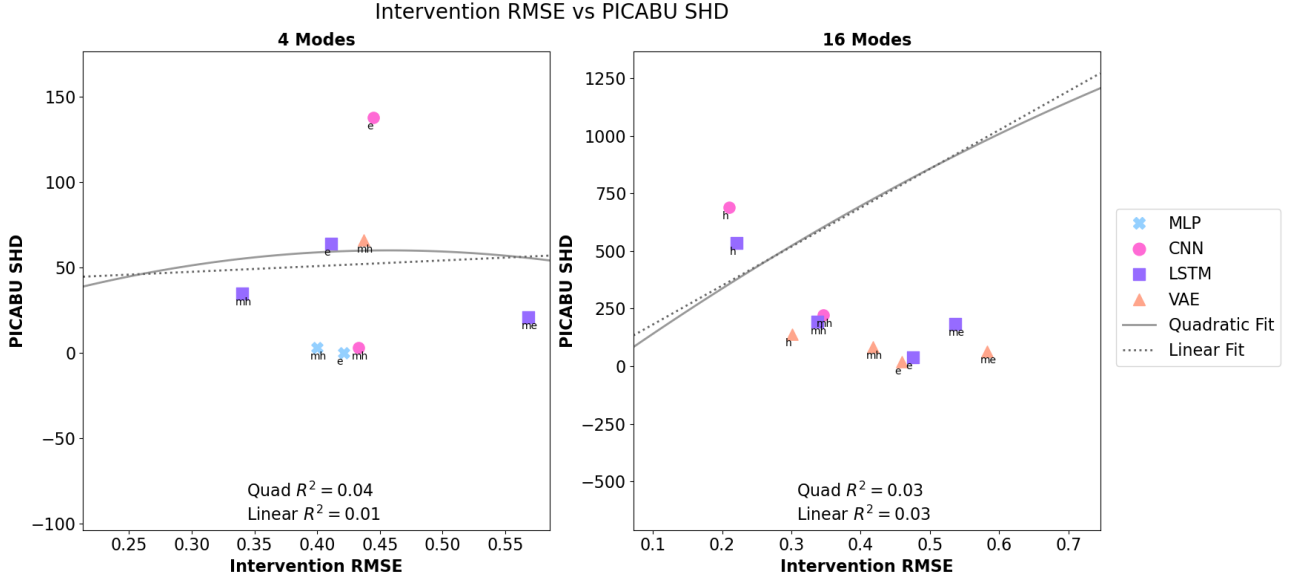


Figure B.15: **Experiment 4: Intervention RMSE vs SHD Score for PICABU.** There no correlation between the structural metric SHD applied to the causal graphs learned by PICABU and the emulator performance metric, Intervention RMSE.

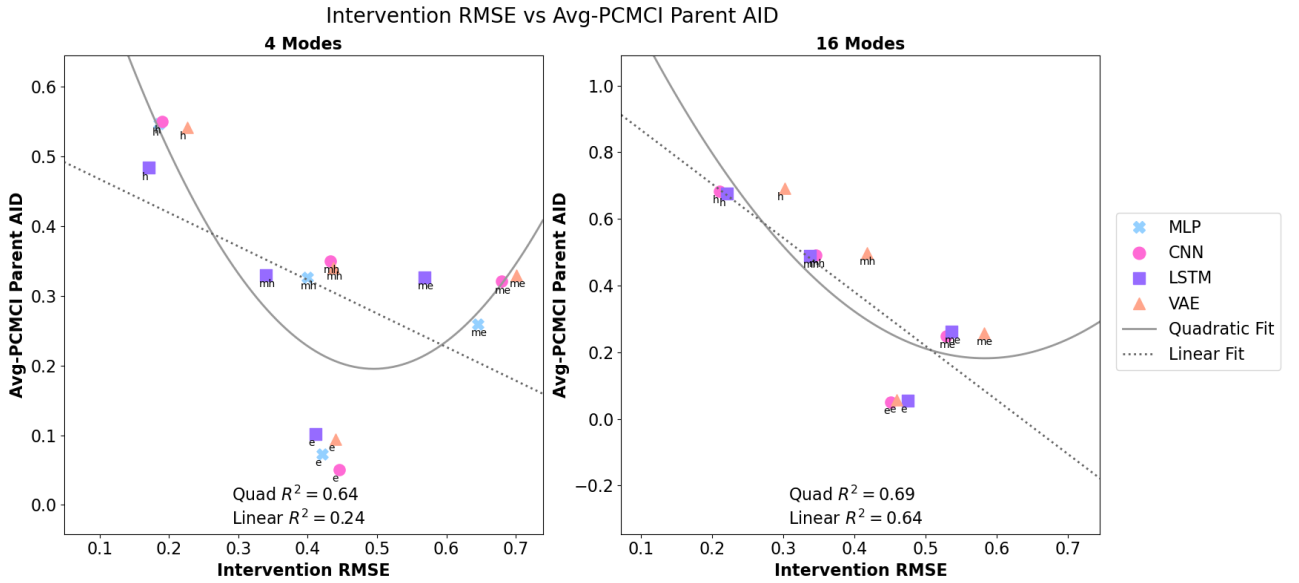


Figure B.16: **Experiment 4: Intervention RMSE vs Parent-AID for Avg-PCMCI.** There is a moderate negative correlation between the causal metric Parent-AID applied to the causal graphs learned by Avg-PCMCI and the emulator performance metric, Intervention RMSE.

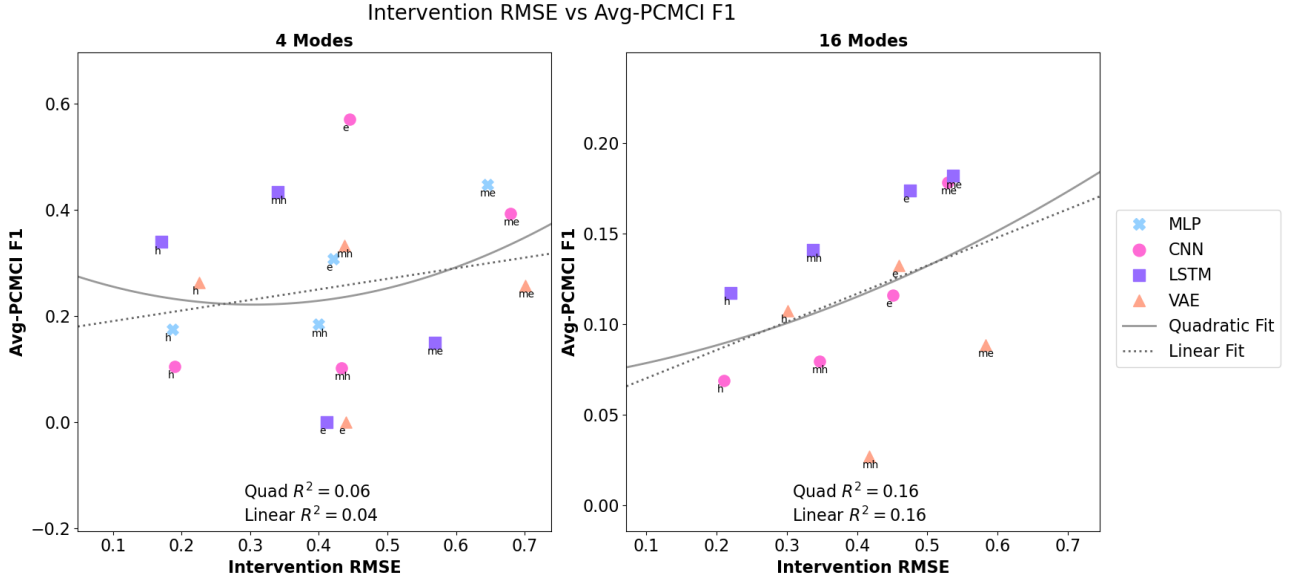


Figure B.17: **Experiment 4: Intervention RMSE vs F1 for Avg-PCMCI.** There is no correlation between the structural metric F1 score applied to the causal graphs learned by Avg-PCMCI and the emulator performance metric, Intervention RMSE.

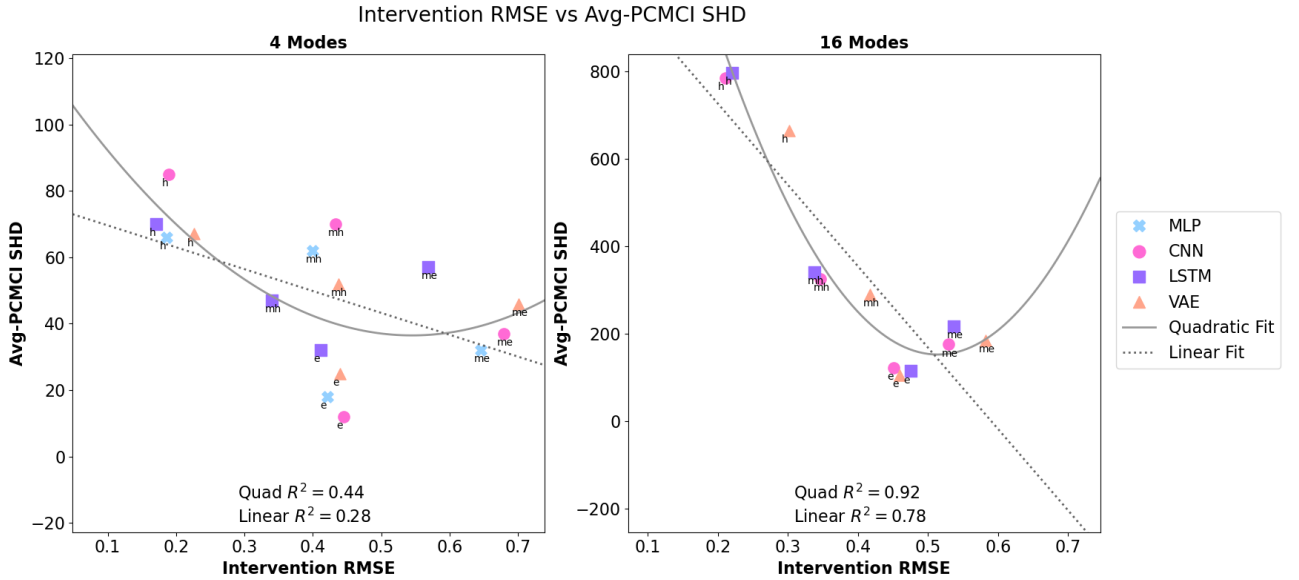


Figure B.18: **Experiment 4: Intervention RMSE vs SHD for Avg-PCMCI.** There is a low-to-moderate negative correlation for 4 modes, and a strong negative correlation for 16 modes between the structural metric SHD score applied to the causal graphs learned by Avg-PCMCI and the emulator performance metric, Intervention RMSE.

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